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Main_Controller IP Objective Specification

STBIO1
Aldo Torino, Sonia Andronaco, Giuliana De Milato

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Draft 1.1	5 October 2025	Aldo Torino, Sonia Andronaco, Giuliana De Milato	
	APPROVED By		DATE
	Francesco Villa, Pasquale Biancolillo		10 October 2025

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1. Introduction

This document is the specification of the Main Controller for STBIO1.

2. Version History

[IPOS_STBIO1_MAIN_CONTROLLER_001] DEFINITION

Current Version	Previous Version	Date	Author	Change description
1.1	1.0	10/09/2026	Alessandro Iannuzzi	Modification on req_id **_7097**
1.0	0.21	10/05/25	Aldo Torino, Sonia Andronaco, Giuliana De Milato	Fix generic typos
0.21	0.20	9/22/2025	Aldo Torino, Sonia Andronaco, Giuliana De Milato	Fix Reqs 7096, 034 and 4034
0.20	0.19	9/18/2025	Aldo Torino, Sonia Andronaco, Giuliana De Milato	Fix Reqs 1131 and 436
0.19	0.18	9/15/2025	Aldo Torino, Sonia Andronaco, Giuliana De Milato	Reference to o_error_time_slot_irq added in req **034.** Added Requirement 4034
0.18	0.17	9/04/2025	Aldo Torino, Sonia Andronaco, Giuliana De Milato	Removed Requirement 1106, changed Requirement 5016, change ALC Requirements and Main Controller Top. Fixed Ports and Parameters table. Added Requirement 1034. Change table in Appendix

Table 1: Version History Table

3. Reference Document

Current Version	Previous Version	Document	Author	Change description
1.11	1.10	AFE_IPOS	Aldo Torino Savino Bellopede	

Table 2: Reference Document Table

4. Terms and Abbreviations

Abbreviation	Meaning
DUT	Device Under Test
HDL	Hardware Description Language
ADC	Analog Digital Converter
FSM	Finite State Machine
ECG	ElectroCardioGraphy
BIA	Bio Impedance
GSR	Galvanic Skin Resistance
PPG	PhotoPlethysmoGram
Bio-Channel	ECG, BIA, GSR
ADSP	Advanced Digital Signal Processing Processor
PMU	Power Management Unit
ISPU	Intelligent Sensor Processor Unit
REGMAP	Register Map

Table 3: Terms and Abbreviations Table

5. Main Controller Top

5.1. Description of Main Controller

The Main Controller is in charge of generating and receiving all the control signals for the Analog Domain, furthermore, performs DSP elaboration on ADC Output and writes it in different Address. This block ~~IP~~ is divided into different sub-blocks, all at the same hierarchy level and described in the next chapter. Together with the Analog Domain, this block is interfaced with the FIFO Controller, the ADSP and the PMU, these last two will not be described in this document. The block ~~IP~~ is divided into different sub-blocks, all at the same hierarchy level and described in the next chapter.

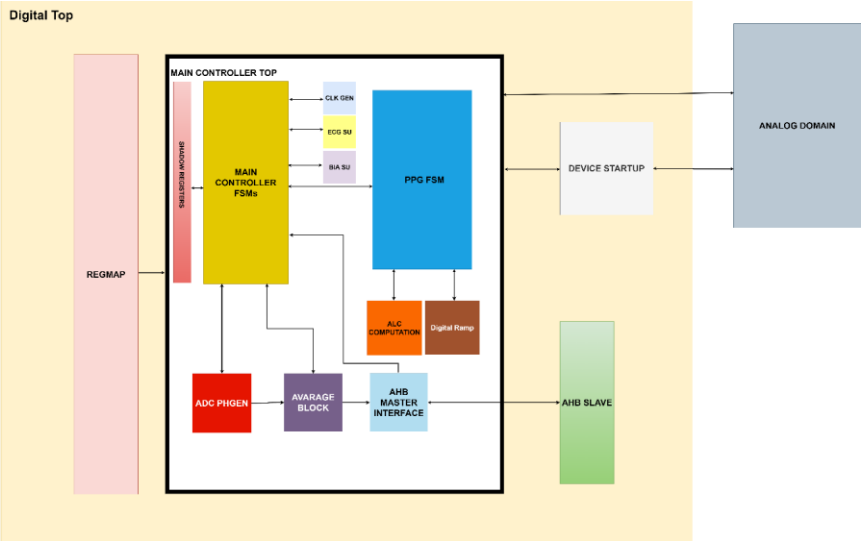


Figure 1: Main Controller Ip Block Diagram

5.1.1. Main controller Top I/O List

5.1.1.1. Parameters

Generic name	Type	Value
ppg_frame_bit_top		4
n_max_frame		12
ppg_rx_channel_bit_top		3
Generic name	**Type**	**Value**

Generic name	Type	Value
ppg_frame_bit_top		4
n_max_frame		12
ppg_rx_channel_bit_top		3
T_tx_led_set_bit_top		5
T_tx_led_off_max_bit_top		7
m_odr_max_bit_top		12
mode_operation_n_bit_top		2
idac_dc_ppg_frame_bit_top		8
time_slot_sel_bit_top		3
T_tx_startup_bit_top		8
T_rx_set_bit_top		9
ppg_ioff_offset_bit_top		10
Nmax_bit_top		4
base_address_bit_top		12
delta_bit_top		2
gain_max_bit_top		4

Generic name	Type	Value
T1_max_bit_top		9
T2_max_bit_top		5
N_clk_max_bit_top		6
n_average_ppg_max_bit_top		3
rx_ch_gain_max_bit_top		4
BTIA_ALC_register_bit_top		9
IDAC_offset		11
n_channel_bit_top		4
n_channel_adc_bit		4
quokka_routine_bit_top		16
vcm_bit_top		14
gnc_cmm_top		15
frame_ppg_debug_alc_top		8
frame_ppg_debug_ppg_raw_top		16
id_frame_max_bit_top		4
data_in_size_bit_top		16

Generic name	Type	Value
ppg_start_up_bit_top		9
ppg_start_up_decod_bit_top		5
data_out_size_bit_top		20
n_average_ecg_bia_gsr_max_bit_top		8
rx_ch_frame_n_bit_top		3
curr_sel_bit_top		8
bit_outab_sel_top		1
n_average_ppg_max_value		128
n_average_ecg_bia_gsr_max_value		16
max_frame_repetition_top		2
n_tx_channel		4

5.1.1.2. Ports

Port name	Direction	Type	Description
i_clk_32kHz	input		clk 32kHz from pmu
i_clk_16MHz	input		clk 16MHz from pmu

Port name	Direction	Type	Description
i_clk_gsr_16M	input		clk for gsr signal
i_clk_ahb_master	input		clock for ahb master
i_rstn	input		reset in 16MHz domain
i_rstn_sync_32	input		reset in 32kz domain
i_wave_velocity_mode	input		enable of the wave velocity mode
i_ispu_end_elab	input		
i_n_ppg_frame	input	[ppg_frame_bit_top-1:0]	number of frame selected
i_const_value_ioff	input	[data_in_size_bit_top-1:0]	constant value from regmap used for ppg ioff calibraton
i_rx_ch_frame	input	[n_max_frame-1:0]	ppg rx channel frame from regamp
i_T_tx_led_set	input	[n_max_frame-1:0]	tx led set time from regmap
i_T_tx_led_off	input	[n_max_frame-1:0]	tx led off time from regmap
i_curr_sel_tx_frame_x	input	[n_max_frame-1:0]	current injection

Port name	Direction	Type	Description
i_tx1_outlab_sel_frame_x	input	[n_max_frame-1:0]	led output selected from regmap for each frame
i_tx2_outlab_sel_frame_x	input	[n_max_frame-1:0]	led output selected from regmap for each frame
i_tx3_outlab_sel_frame_x	input	[n_max_frame-1:0]	led output selected from regmap for each frame
i_tx4_outlab_sel_frame_x	input	[n_max_frame-1:0]	led output selected from regmap for each frame
i_idac_dc_ppg_frame	input	[n_max_frame-1:0]	idac value for each frame selected from regmap
i_average_ppg	input	[n_max_frame-1:0]	average ppg for each frame selected from regmap
i_gain_alc_start	input	[gain_max_bit_top-1:0]	btia gain for alc from regmap
i_gain_start	input	[n_max_frame-1:0]	btia gain start for compensation from regmap
i_Tx_idac_Vref_debug	input	[Nmax_bit_top-1:0]	Idac value for debug from regmap

Port name	Direction	Type	Description
i_ppg_start_up_time	input	[ppg_start_up_decod_bit_top-1:0]	start up time for ppg from regmap
i_en_txi	input	[n_max_frame-1:0]	tx enabling for each frame from regmap
i_frame_repetition	input	[n_max_frame-1:0]	i frame repetition for each frame from regmap
i_T_preset	input	[1:0]	Preset time for digital ramp from regmap
i_m_ecg	input	[m_odr_max_bit_top-1:0]	m parameter for ecg from regmap
i_m_bia	input	[m_odr_max_bit_top-1:0]	m parameter for bia from regmap
i_m_gsr	input	[m_odr_max_bit_top-1:0]	m parameter for gsr from regmap
i_m_ppg	input	[m_odr_max_bit_top-1:0]	m parameter for ppg from regmap
i_debug_adc_error_cap8_0_bypass	input		error cap bypass for adc from regmap
i_base_address_ecg0	input	[base_address_bit_top-1:0]	base address for ecg0 in multiple fifo
i_base_address_bia	input	[base_address_bit_top-1:0]	base address for bia/ecg1-2 in multiple fifo

Port name	Direction	Type	Description
i_base_address_gsr	input	[base_address_bit_top-1:0]	base address for gsr in multiple fifo
i_base_address_ppg	input	[base_address_bit_top-1:0]	base address for ppg in multiple fifo
i_frame_ppg_debug	input	[ppg_frame_bit_top-1:0]	frame selection for ppg data debug
i_en_buf_gpio1	input		enable buffer on gpio1
i_en_buf_gpio2	input		enable buffer on gpio2
i_en_biobuf	input		enable biobuffer
i_do_calibration	input		enabling for adc calibration
i_mode_operation	input	[mode_operation_n_bit_top-1:0]	mode operation selection
i_T_rx_set	input	[T1_max_bit_from_decoder_top-1:0]	rx channel time set
i_en_clkf	input		en 16 MHz clock
i_quokka_boot_end	input		from quokka
i_quokka_run	input		from quokka
i_clk_16MHz_ready	input		(from analog circuit)

Port name	Direction	Type	Description
i_imp_curinj_en	input		enabling current injection for bia
i_en_ldo_1v8	input		enabling ldo 1v8
i_debug_comp_rx_ck	input		enabling debug for comp rx ck
i_debug_gsr_on	input		gsr on debug mode
i_debug_gsr_curr_on	input		gsr curr on debug mode
i_alc_disabled	input	[n_max_frame-1:0]	disabling alc bit for each frame
i_en_cm_buf	input		enabling common buffer
i_en_rld	input		enabling rld
i_en_avg1	input		enabling avg1
i_en_avg2	input		enabling avg1
i_hc_en	input		enabling health channel
i_debug	input		enabling debug mode
i_imp_en	input		enabling bia channel

Port name	Direction	Type	Description
i_debug_en_tia	input		enabling debug tia
i_debug_en_buffer	input		enabling debug for buffer
i_debug_en_idac	input		enabling debug for IDAC
i_en_clk_chop10k	input		enabling chopper clock
i_debug_tx_preset	input		enabling tx preset
i_en_ldo_1v8_ext	input		enabling external ldo 1v8
i_clk_imp_10k_en	input		enabling clock chopper for bia
i_b_cap_idac	input	[1:0]	enabling cap for idac
i_adc_mux_bio_channel_debug	input	[2:0]	enabling debug adc mux bio channel
i_adc_mux_sel_debug	input		enabling adc mux
i_ispu_core_sw_rstn	input		ispu core enabling
i_T_TIAON	input	[T1_max_bit_from_decoder_top-1:0]	TIA Time
i_scan_mode	input		scan mode enabling

Port name	Direction	Type	Description
i_reset_debug_adc_data	input		reset adc data debug
i_start_adc_debug	input		pilot adc start om debug
i_en_ADC_debug	input		enabling adc debug
i_debug_adc_readout_bypass	input		enabling adc readout bypass
i_su_hlt_dly	input		enabling start up delay for ecg
i_multiple_fifo_sel	input		enabling multiple fifo
i_debug_calibration	input		enabling debug calibraiton
i_en_gsr_dc_cds	input	[1:0]	enabling gsr cds mode
i_ADC_EOC	input		end of conversion from adc
i_ecg0_sel	input		select ecg0 channel
i_bia_sel	input		select bia channel
i_gsr_sel	input		select gsr channel
i_ppg_sel	input		select ppg channel

Port name	Direction	Type	Description
i_ppg_ioff_calibration	input		enabling ppg ioff calibration
i_ecg1_ecg2_sel	input		enabling ecg1 and ecg2 channel
i_rst_async_ecg	input		reset async for ecg
i_rst_async_bia	input		reset async for bia
i_ppg_quokka_elab_sel	input		enabling adsp elab for ppg
i_data_ADC_out	input	[data_in_size_bit_top-1:0]	(from analog circuit)
i_time_slot_selection	input	[time_slot_sel_bit_top-1:0]	odr selection from regmap
i_Nmax_to_reg	input	[Nmax_bit_top-1:0]	Max value for digital ramp
i_nstart_case	input	[Nmax_bit_top-1:0]	n_start value for frist sterp
i_debug_rx_ch	input	[rx_ch_frame_n_bit_top-1:0]	enabling debug for rx channel
i_T_delta1_to_reg	input	[delta_bit_top-1:0]	delta 1 time for digital ramp
i_gain_max_set	input	[gain_max_bit_top-1:0]	gain max for alc
i_BTIA_ALC_debug	input	[BTIA_ALC_register_bit_top-1:0]	BTIA ALC Debug enable

Port name	Direction	Type	Description
i_comp_rx_out	input		input from analog comparator
i_adc_error_cap8_debug	input	[6:0]	adc cap error debug
i_adc_error_cap7_debug	input	[6:0]	adc cap error debug
i_adc_error_cap6_debug	input	[6:0]	adc cap error debug
i_adc_error_cap5_debug	input	[6:0]	adc cap error debug
i_adc_error_cap4_debug	input	[6:0]	adc cap error debug
i_adc_error_cap3_debug	input	[6:0]	adc cap error debug
i_adc_error_cap2_debug	input	[6:0]	adc cap error debug
i_adc_error_cap1_debug	input	[6:0]	adc cap error debug
i_adc_error_cap0_debug	input	[6:0]	adc cap error debug
i_BIDAC_debug_register	input	[IDAC_offset-1:0]	
i_sleep_master	input		

Port name	Direction	Type	Description
i_lpm	input	[1:0]	low power mode for ldo
i_mpm	input	[1:0]	medium power mode for ldo
i_hpm	input	[1:0]	high power mode for ldo
i_dis_rc_viref	input		dis rc viref
i_T1	input	[T1_max_bit_from_decoder_top-1:0]	4 to 10 us=160nclk
i_T2	input	[T2_max_bit_from_decoder_top-1:0]	0.5 to 2 us=~32
i_N_clk_ppg	input	[N_clk_max_bit_top-1:0]	Sample time for ppg
i_N_clk_ecg	input	[N_clk_max_bit_top-1:0]	Sample time for ecg
i_N_clk_cal	input	[N_clk_max_bit_top-1:0]	Sample time for adc calibration
scan_enable	input		scan enable for scan mode
i_timer_buffer_config	input	[1:0]	timer buffer configuration
i_ppg_alc_frame_save	input	[ppg_frame_bit_top-1:0]	alc data frame to save
i_reset_ppg_ioff_calib	input		

Port name	Direction	Type	Description
HRDATA	input	[31:0]	AHB Protocol signals
HREADY	input		AHB Protocol signals
HRESP	input	[1:0]	AHB Protocol signals
HGRANT	input		AHB Protocol signals
SCANENABLE	input		Scan Test Mode Enbl
SCANINHCLK	input		Scan Chain Input
SCANOUTHCLK	output		Scan Chain Output
HSEL	output		AHB Protocol signals
HADDR	output	[31:0]	AHB Protocol signals
HTRANS	output	[1:0]	AHB Protocol signals
HWRITE	output		AHB Protocol signals
HSIZE	output	[2:0]	AHB Protocol signals

Port name	Direction	Type	Description
HBURST	output	[2:0]	AHB Protocol signals
HPROT	output	[3:0]	AHB Protocol signals
HWDATA	output	[31:0]	AHB Protocol signals
HBUSREQ	output		AHB Protocol signals
HLOCK	output		AHB Protocol signals
o_bio_clock_div	output		output for clock chopper
o_BIDAC	output	[IDAC_offset-1:0]	IDAC for top
o_rst_imp	output		bia start up rst signal
o_rsti_imp	output		bia start up rsti signal
o_sug_imp	output		bia start up sug channel
o_pd_imp	output		bia start up pd channel
o_start_operative_16MHz	output		start operative 16MHz

Port name	Direction	Type	Description
o_su_hc2	output		ecg start up hc2 signal
o_rst_hc2	output		ecg start up rst signal
o_rsti_hc2	output		ecg start up rsti signal
o_pd_hc2	output		ecg start up pd signal
o_reset_reg_rst	output		end of reset async phase
o_reset_reg_rst_imp_async	output		end of reset async phase
o_imp_curinj_pd	output		bia current injection output
o_Txi_IDAC_Vref	output	[Nmax_bit_top-1:0]	TX IDAC VREF TO analog domain
o_comp_rx_ck	output		output for comparator debug
o_BTIA_gain	output	[gain_max_bit_top-1:0]	BTIA gain output to analog domain
o_data_out_debug_adc	output	[15:0]	o data output for adc debug
o_en_mask_ldo_pdclkf_reg	output		mask for en_ldo and pdf

Port name	Direction	Type	Description
o_pd_clkf	output		pd_clkf
o_trim_ldo_d	output	[1:0]	trim ldo value
o_BDC_PPG	output	[idac_dc_ppg_frame_bit_top-1:0]	idac dc ppg frame output
o_adc_mux_bio_channel	output	[2:0]	adc mux bio channel
o_adc_mux_sel	output		adc mux sel output
o_error_time_slot	output		error time slot irq
o_curr_sel_tx_frame_x	output	[curr_sel_bit_top-1:0]	current selection for tx channel output
o_tx1_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	tx outlab sel frame x
o_tx2_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	tx outlab sel frame x
o_tx3_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	tx outlab sel frame x
o_tx4_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	tx outlab sel frame x
o_tx_preset	output		tx prese to analog domain
o_en_tx	output	[n_tx_channel-1:0]	tx enable to analog domain

Port name	Direction	Type	Description
o_PD_TIA	output		enabling tia to analog domain
o_PD_PGA	output		enabling buffer to analog domain
o_PD_IDAC	output		enabling idac to analog domain
o_gsr_curr_on	output		gsr curr on pilot to analog domain
o_gsr_on	output		grs on on pilot to analog domain
o_ADC_en	output		ADC enabling
o_half_charge_ldo	output		
o_ADC_start	output		ADC Start sampling
o_N_step2_to_reg	output	[rx_ch_gain_limit-1:0]	N_Step2 calibration value
o_adc_test_cap	output	[9:0]	adc test cap
o_adc_error_cap9	output	[7:0]	adc calibration result
o_adc_error_cap8	output	[6:0]	adc calibration result
o_adc_error_cap7	output	[6:0]	adc calibration result

Port name	Direction	Type	Description
o_adc_error_cap6	output	[6:0]	adc calibration result
o_adc_error_cap5	output	[6:0]	adc calibration result
o_adc_error_cap4	output	[6:0]	adc calibration result
o_adc_error_cap3	output	[6:0]	adc calibration result
o_adc_error_cap2	output	[6:0]	adc calibration result
o_adc_error_cap1	output	[6:0]	adc calibration result
o_adc_error_cap0	output	[6:0]	adc calibration result
o_pd_cm_buf	output		power down common buffer
o_pd_rld	output		power down rld
o_pd_avg1	output		power down avg1
o_pd_avg2	output		power down avg2
o_mc_ahb_clk_en	output		enable for master ahb clock
o_main_controller_device_status_register	output	[2:0]	device status register

Port name	Direction	Type	Description
o_main_controller_elab_status_register	output	[3:0]	device status register
o_en_ldo_1v8	output		enable ldo
o_en_bufxbio	output		enable biobuffer to analog domain
o_rx_ch_frame	output	[ppg_rx_channel_bit_top-1:0]	selected rx ch frame to analog domain
o_ldo_ext	output		enable external ldo
o_ppg_status_register	output	[n_state_bit-1:0]	ppg state status register
o_b_cap_idac	output	[1:0]	b cap idac to analog domain
o_ispu_irq_ctrl	output	[3:0]	ispu irq
o_irq_start_measurement	output		irq start measurement to GPIO
o_end_calibration	output		end calibration to analog domain
o_alc_data	output	[data_out_size_bit_top+2:0]	alc data to regmap
o_ppg_data_raw	output	[data_out_size_bit_top+2:0]	ppg raw data to regmap
o_pd_buf_gpio1	output		power down for buffer gpio1

Port name	Direction	Type	Description
o_pd_buf_gpio2	output		power down for buffer gpio2
o_adc_data_debug_alc_1	output	[frame_ppg_debug_alc_top-1:0]	alc 1 registers data debug
o_adc_data_debug_alc_2	output	[frame_ppg_debug_alc_top-1:0]	alc 2 registers data debug
o_adc_data_debug_ppg_raw	output	[frame_ppg_debug_ppg_raw_top-1:0]	ppg raw registers data debug
o_en_lead_off	output		mask lead off
o_mc_busy	output		main controller is working
o_en_ck_m_p_q	output		clock m and q bia
o_en_ck_chop_imp	output		clock chopper bia
o_saturation_flag	output	[n_max_frame-1:0]	ppg saturation flag
o_en_ck_hc2_10k	output		clock chipper ecg
o_en_clk_16MHz_sleep	output		sleep signal for pmu

Table 4: I/O LIst Main Controller Top

5.1.1.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_3003] ASSUMPTION:
The following channels combination are the only ones supported by STBIO1:

1. 1 ECG channel (ECG0)
2. 2 ECG channels (ECG1 + ECG2)
3. 3 ECG channels (ECG0 + ECG1 + ECG3)
4. BIA channel
5. GSR channel
6. PPG channel
7. ECG0 and BIA channels
8. ECG0 and GSR channels
9. ECG0 and PPG channels
10. ECG1 and ECG2 and PPG channels
11. BIA and PPG channels
12. GSR and PPG channels
13. ECG0 and ECG1 and ECG2 and PPG channels
14. ECG0 and BIA and PPG channels
15. ECG0 and GSR and PPG channels

[END]

5.2. Configuration Register Update

5.2.1. I/O List

5.2.1.1. Parameters

Generic name	Type	Value	Description
m_odr_max_bit	int	11	
mode_operation_n_bit	int	2	
time_slot_sel_bit	int	3	
N_clk_max_bit	int	7	
quokka_routine_bit	int	3	

Generic name	Type	Value	Description
T_tx_startup_bit	int	5	
T_rx_set_bit	int	5	
ppg_rx_channel_bit	int	3	
T_tx_led_set_bit	int	5	
T_tx_led_off_max_bit	int	8	
idac_dc_ppg_frame_bit	int	7	
n_average_ecg_bia_gsr_max_bit	int	4	
n_channel_bit	int	4	
n_average_ppg_max_bit	int	4	
max_frame_repetition	int	2	
curr_sel_bit	int	9	
bit_outab_sel	int	1	
n_tx_channel	int	4	
n_channel_adc_bit	int	4	
ecg_channel_bit	int	2	
time_buffer_const_bit	int	10	
Nmax_bit	int	4	

Generic name	Type	Value	Description
delta_bit	int	2	
gain_max_bit	int	3	
T1_max_bit_from_decoder	int	4	
T2_max_bit_from_decoder	int	2	
T1_max_bit	int	8	
T2_max_bit	int	5	
BTIA_ALC_register_bit	int	10	
T1_add_T2_max_bit	int	13	
ppg_start_up_decod_bit	int	4	
ppg_frame_bit	int	4	
ppg_frame_bit_ALC_BTIA	int	12	
n_max_frame	int	5	

5.2.1.2. Ports

Port name	Direction	Type	Description
i_clk	input		
i_rst_n	input		
i_en_cm_buf	input		

Port name	Direction	Type	Description
i_en_rld	input		
i_en_avg1	input		
i_en_avg2	input		
i_en_gsr_dc_cds	input	[1:0]	
i_hc_en	input		
i_start_su_ecg_mc	input		
i_start_su_bia_mc	input		
i_debug	input		
i_imp_en	input		
i_imp_curinj_en	input		
i_ppg_quokka_elab_sel	input		
i_idac_dc_ppg_frame	input	[n_max_frame-1:0][idac_dc_ppg_frame_bit-1:0]	
i_check_update	input		
i_reset_su	input		
i_cfg_enable	input		
i_Nmax_to_reg	input	[Nmax_bit-1:0]	

Port name	Direction	Type	Description
i_time_slot_selection	input	[time_slot_sel_bit-1:0]	
i_ppg_start_up_time	input	[ppg_start_up_decod_bit-1:0]	
i_first_cfg	input		
i_gain_max	input	[gain_max_bit-1:0]	
i_n_ppg_frame	input	[ppg_frame_bit-1:0]	
i_rx_ch_frame	input	[n_max_frame-1:0][ppg_rx_channel_bit-1:0]	
i_T_tx_led_set	input	[n_max_frame-1:0][T_tx_led_set_bit-1:0]	
i_T_tx_led_off	input	[n_max_frame-1:0] [T_tx_led_off_max_bit-1:0]	
i_curr_sel_tx_frame_x	input	[n_max_frame-1:0][curr_sel_bit-1:0]	
i_tx1_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	
i_tx2_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	
i_tx3_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	
i_tx4_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	
i_average_ppg	input	[n_max_frame-1:0][n_average_ppg_max_bit-1:0]	
i_gain_start	input	[n_max_frame-1:0][gain_max_bit-1:0]	
i_en_txi	input	[n_max_frame-1:0][3:0]	

Port name	Direction	Type	Description
i_gain_alc_start	input	[gain_max_bit-1:0]	
i_en_ldo_1v8_ext	input		
i_frame_repetition	input	[n_max_frame-1:0][max_frame_repetition-1:0]	
i_T_rx_set	input	[T1_max_bit_from_decoder-1:0]	
i_nstart_case	input	[Nmax_bit-1:0]	
i_T_delta1_to_reg	input	[delta_bit-1:0]	
i_wave_velocity_mode	input		
i_b_cap_idac	input	[1:0]	
i_T1	input	[T1_max_bit_from_decoder-1:0]	
i_T2	input	[T2_max_bit_from_decoder-1:0]	
i_T_TIAON	input	[T1_max_bit_from_decoder-1:0]	TIA Time
o_pd_cm_buf	output		
o_pd_rld	output		
o_pd_avg1	output		
o_pd_avg2	output		
o_hc_en	output		

Port name	Direction	Type	Description
o_idac_dc_ppg_frame_cfg	output	[n_max_frame-1:0]	
o_imp_en	output		
o_imp_curinj_en	output		
o_ppg_quokka_elab_sel	output		
o_n_ppg_frame_cfg	output	[ppg_frame_bit-1:0]	
o_rx_ch_frame_cfg	output	[n_max_frame-1:0]	
o_T_tx_led_set_cfg	output	[n_max_frame-1:0]	
o_T_tx_led_off_cfg	output	[n_max_frame-1:0]	
o_curr_sel_tx_frame_x_cfg	output	[n_max_frame-1:0]	
o_tx1_outlab_sel_frame_x_cfg	output	[n_max_frame-1:0]	
o_tx2_outlab_sel_frame_x_cfg	output	[n_max_frame-1:0]	
o_tx3_outlab_sel_frame_x_cfg	output	[n_max_frame-1:0]	
o_tx4_outlab_sel_frame_x_cfg	output	[n_max_frame-1:0]	
o_average_ppg_cfg	output	[n_max_frame-1:0]	
o_gain_start_cfg	output	[n_max_frame-1:0]	
o_gain_max_cfg	output	[gain_max_bit-1:0]	

Port name	Direction	Type	Description
o_en_txi_cfg	output	[n_max_frame-1:0]	
o_frame_repetition_cfg	output	[n_max_frame-1:0]	
o_T_rx_set_cfg	output	[T1_max_bit_from_decoder-1:0]	
o_nstart_case_cfg	output	[Nmax_bit-1:0]	
o_T_delta1_to_reg_cfg	output	[delta_bit-1:0]	
o_Nmax_to_reg_cfg	output	[Nmax_bit-1:0]	
o_T1_cfg	output	[T1_max_bit_from_decoder-1:0]	
o_T2_cfg	output	[T2_max_bit_from_decoder-1:0]	
o_ppg_start_up_time	output	[ppg_start_up_decod_bit-1:0]	
o_wave_velocity_mode_cfg	output		
o_time_slot_selection_cfg	output	[time_slot_sel_bit-1:0]	
o_en_gsr_cds	output		
o_gsr_dc_mode	output		
o_T_TIAON_cfg	output	[T1_max_bit_from_decoder-1:0]	TIA Time
o_gain_alc_start	output	[gain_max_bit-1:0]	
o_en_ldo_1v8_ext	output		

Port name	Direction	Type	Description
o_reset_time_slot_cfg	output		
o_done_cfg	output		
o_change_curr	output		
o_change_gain_start	output		
o_change_idac_dc_ppg	output		
o_b_cap_idac	output	[1:0]	

Table 5:Configuration Register Update

5.2.2. Description of the Block

This block oversees the correct configuration of register useful to configure BIO-Channel or PPG Measurement and avoids that run-time changes from the user during the measurement.
The description of each input signal is written in the linked document in the next paragraph. All the registers are regmap shadow registers.

5.2.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_7096] REQUIREMENT
When the device FSM is in OPERATIVE or SLEEP state and the ppg channel is activated, if the user changes i_curr_sel_tx_frame_x, i_gain_start or i_idac_dc_ppg_frame, the signals o_change_curr, o_change_gain_start or o_change_idac_dc_ppg (respectively) shall go high at the start of the next time slot (when passing from Sleep to Operative State) and shall stay high until another one of these three value is changed.
[END]

[IPOS_STBIO1_MAIN_CONTROLLER_7097] REQUIREMENT
If i_cfg_enable and i_first_cfg is set to 1, o_change_curr, o_change_gain_start or o_change_idac_dc_ppg shall go to 0 in less than 4 clock cycles
[END]

Commented [AI1]: I do a light modification to test the deterministic workflow: from 5 to 4 clock cycles

[IPOS_STBIO1_MAIN_CONTROLLER_7098] REQUIREMENT

If one or more signals between o_change_curr, o_change_gain_start or o_change_idac_dc_ppg is set to 1, a data with the following format {21'hFFFFE, o_change_curr, o_change_gain_start, o_change_idac_dc_ppg} shall be written in FIFO before the next time slot Data. If the FIFO is in Multiple FIFO, the Data shall be written in the Sub-FIFO selected for PPG by the user
[END]

[IPOS_STBIO1_MAIN_CONTROLLER_002] REQUIREMENT:
The only registers that the user shall change after setting the mode_operation to 1 are the ones described in requirement 7096. All the Other signals shall change only with mode_operation set to 0 or when i_debug is set to 1.**[END]**

5.3. Main Controller FSMs

5.3.1. I/O List

5.3.1.1. Parameters

Generic name	Type	Value	Description
ppg_frame_bit	int	4	
m_odr_max_bit	int	11	
time_slot_sel_bit	int	3	
N_clk_max_bit	int	7	
quokka_routine_bit	int	3	
ppg_rx_channel_bit	int	3	
T_tx_led_set_bit	int	5	
T_tx_led_off_max_bit	int	8	
idac_dc_ppg_frame_bit	int	7	

Generic name	Type	Value	Description
n_average_ecg_bia_gsr_max_bit	int	4	
n_channel_bit	int	4	
n_average_ppg_max_bit	int	4	
n_max_frame	int	5	
rx_ch_gain_max_bit	int	4	
curr_sel_bit	int	9	
bit_outab_sel	int	1	
n_tx_channel	int	4	
base_address_bit	int	12	
n_channel_adc_bit	int	4	
ahb_base_address	logic [31:0]	32'h000C_4000	
ecg_channel_bit	int	2	
time_buffer_const_bit	int	10	

5.3.1.2. Ports

Port name	Direction	Type	Descr.
i_clk_32Khz	input		clock 32kHz

Port name	Direction	Type	Descr.
i_clk_16MHz	input		clock 16MHz
i_clk_16MHz_write	input		ahb clock
i_clk_elab_fsm	input		elab fsm clock
i_clk_boot_32k	input		32k clock boot
i_clk_gsr_16M	input		clock for gsr
i_clk_change_address	input		clock for change address
i_time_slot_count_clk	input		time slot count clock
i_rstn	input		reset 16MHz
i_rstn_sync_32	input		reset 32 kHz
i_n_ppg_frame	input	[ppg_frame_bit-1:0]	number of frame for ppg
i_rx_ch_frame	input	[n_max_frame-1:0][ppg_rx_channel_bit-1:0]	rx channel for each ppg frame
i_average_ppg	input	[n_max_frame-1:0][n_average_ppg_max_bit-1:0]	average for each ppg frame

Port name	Direction	Type	Descr.
i_T_tx_led_set	input	[n_max_frame-1:0][T_tx_led_set_bit-1:0]	time tx led set for each frame
i_ppg_ioff_calibration	input		ioff calibration mode for ppg
i_end_ioff_calibration	input		end ioff calibration mode for ppg
i_ppg_alc_frame_save	input	[ppg_frame_bit-1:0]	ppg alc frame save
i_T_tx_led_off	input	[n_max_frame-1:0][T_tx_led_off_max_bit-1:0]	time led off for each frame
i_curr_sel_tx_frame_x	input	[n_max_frame-1:0][curr_sel_bit-1:0]	curr sel tx for each frame
i_tx1_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	tx outlab sel for each frame and tx
i_tx2_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	tx outlab sel for each frame and tx
i_tx3_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	tx outlab sel for each frame and tx

Port name	Direction	Type	Descr.
i_tx4_outlab_sel_frame_x	input	[n_max_frame-1:0][bit_outab_sel-1:0]	tx outlab sel for each frame and tx
i_idac_dc_ppg_frame	input	[n_max_frame-1:0][idac_dc_ppg_frame_bit-1:0]	idac dc ppg frame for each frame
i_en_ldo_1v8_ext	input		enable ldo ext
i_debug	input		enable debug mode
i_debug_gsr_on	input		enable debug for gsr
i_ispu_end_elab	input		end ispu elaboration
i_gsr_dc_mode	input		gsr dc mode
i_en_tx_ch	input	[n_tx_channel-1:0]	ppg tx channel enable
i_debug_gsr_curr_on	input		debug gsr curr on value
i_multiple_fifo_sel	input		multiple fifo channel mode
i_n_ecg_channel	input	[ecg_channel_bit-1:0]	ecg active channel

Port name	Direction	Type	Descr.
i_N_clk_cal	input	[N_clk_max_bit-1:0]	sample clock cycles for calibration
i_ispu_core_sw_rstn	input		enable for ispu power island
i_timer_buffer_config	input	[1:0]	time buffer value
i_ppg_quokka_elab_sel	input		adsp data elab mode
i_lpm	input	[1:0]	low power mode for ldo
i_mpm	input	[1:0]	medium power mode for ldo
i_hpm	input	[1:0]	high power mode for ldo
i_rst_async_ecg	input		rst async for ecg
i_alc_disabled	input	[n_max_frame-1:0]	alc disabled for ppg
i_rst_async_bia	input		rst async for bia

Port name	Direction	Type	Descr.
i_done_cfg	input		end of configuration
i_scan_mode	input		scan mode enable
i_m_ecg	input	[m_odr_max_bit-1:0]	division factor for ecg
i_m_bia	input	[m_odr_max_bit-1:0]	division factor for bia
i_m_gsr	input	[m_odr_max_bit-1:0]	division factor for gsr
i_m_ppg	input	[m_odr_max_bit-1:0]	division factor for ppg
i_start_operation	input		start operation
i_base_address_ecg0	input	[base_address_bit-1:0]	base address multiple fifo for ecg0
i_base_address_bia	input	[base_address_bit-1:0]	base address multiple fifo for bia/ecg1&2
i_base_address_gsr	input	[base_address_bit-1:0]	base address multiple fifo for gsr

Port name	Direction	Type	Descr.
i_base_address_ppg	input	[base_address_bit-1:0]	base address multiple fifo for ppg
i_bia_sel	input		bia channel selection
i_gsr_sel	input		gsr channel selection
i_ppg_sel	input		ppg channel selection
i_dis_rc_viref	input		dis rc viref from pmu
i_change_cfg_ppg	input		change configuration signal for ppg
i_change_curr	input		change curr for ppg
i_change_gain_start	input		change gain start for ppg
i_change_idac_dc_ppg	input		change idac value for ppg
i_sleep_master	input		sleep master from sensor hub

Port name	Direction	Type	Descr.
i_adc_mux_bio_channel_debug	input	[2:0]	adc mux bio channel
i_adc_mux_sel_debug	input		adc mux debug
i_quokka_boot_end	input		adsp boot end signal
i_data_storage_mode	input		data storage mode
i_quokka_run	input		adsp elaboration running
i_end_of_conversion	input		end of converison
i_end_of_ppg	input		end of ppg
i_clk_16MHz_ready	input		ready from 16 MHz ring oscillator
i_hc_su	input		health channel start up end
i_end_calibration	input		end adc calibration
i_imp_su	input		bia start up phase end

Port name	Direction	Type	Descr.
i_write_done	input		write done from ahb
i_time_slot_selection	input	[time_slot_sel_bit-1:0]	time slot selection
i_N_clk_ppg	input	[N_clk_max_bit-1:0]	ppg sampling phase clock cycles
i_N_clk_ecg	input	[N_clk_max_bit-1:0]	ecg sampling phase clock cycles
i_adc_ready	input		adc ready
i_busy_ppg	input		ppg running
i_en_gsr_cds	input		en gsr cds mode
i_do_calibration	input		do calibration for adc
i_reset_curr_frame	input		reset current ppg frame
i_wave_velocity_mode	input		enable wave velocity mode
i_start_up_time_ppg_done	input		start up ppg done

Port name	Direction	Type	Descr.
i_reset_async_reg_ecg	input		reset async ecg done
i_reset_async_reg_bia	input		reset async bia done
o_ADC_en_mc	output		adc enable main controller
o_start_adc	output		start for analog adc
o_start_ppg	output		start ppg fsm
o_pwr_dwn_16MHz	output		enable for 16MHz ring oscillator
o_reset_average	output		reset average
o_quokka_data_valid	output	[quokka_routine_bit-1:0]	start ADSP
o_n_ppg_frame	output	[ppg_frame_bit-1:0]	number for ppg frame
o_rx_ch_frame	output	[ppg_rx_channel_bit-1:0]	rx channel output
o_en_ldo_1v8	output		enable ldo 1v8
o_T_tx_led_set	output	[T_tx_led_set_bit-1:0]	time led set for ppg fsm

Port name	Direction	Type	Descr.
o_T_tx_led_off	output	[T_tx_led_off_max_bit-1:0]	time led off for ppg fsm
o_idac_dc_ppg_frame	output	[idac_dc_ppg_frame_bit-1:0]	idac dc ppg for ppg fsm
o_curr_sel_tx_frame_x	output	[curr_sel_bit-1:0]	curr sel tx for frame running
o_tx1_outlab_sel_frame_x	output	[bit_outab_sel-1:0]	tx outab sel
o_tx2_outlab_sel_frame_x	output	[bit_outab_sel-1:0]	tx outab sel
o_tx3_outlab_sel_frame_x	output	[bit_outab_sel-1:0]	tx outab sel
o_tx4_outlab_sel_frame_x	output	[bit_outab_sel-1:0]	tx outab sel
o_n_average	output	[n_average_ecg_bia_gsr_max_bit-1:0]	average for channel on measuring
o_adc_mux_bio_channel	output	[2:0]	adc mux to analog domain
o_adc_mux_sel	output		adc mux selection to analog domain
o_N_clk	output	[N_clk_max_bit-1:0]	clock cycles for adc sampling

Port name	Direction	Type	Descr.
o_data_valid_ecg_bia_adc	output		data valid for ecg and bia
o_average_ppg	output	[n_average_ppg_max_bit-1:0]	average for ppg channel
o_error_time_slot	output		error irq
o_ahb_address	output	[31:0]	address for ahb write operation
o_change_address_clk_en	output		change address clock gating enable
o_reset_write	output		reset write operation
o_done_frame	output		end frame ppg
o_average_clk_en	output		average ip clock gating enable
o_ppg_fsm_clk_en	output		ppg fsm ip clock gating enable
o_quokka_start_en	output		adsp start enable

Port name	Direction	Type	Descr.
o_trim_ldo_d	output	[1:0]	trim ldo output
o_calibration_phase	output		end of calibration phase
o_quokka_write	output		write ahb for adsp
o_tag_data	output	[7:0]	data tag
o_write_data_tag_ahb_en	output	[1:0]	write data tag ahb
o_gsr_curr_on	output		gsr curr on signal to analog domain
o_gsr_on	output		gsr on signal to analog domain
o_gsr_result	output		gsr result computation to average ip
o_data_valid_gsr_adc	output		data valid gsr adc
o_reset_calibration_phase	output		reset adc calibration phase

Port name	Direction	Type	Descr.
o_en_clk_16MHz_sleep	output		close clock in sleep state
o_mc_ahb_clk_en	output		master ahb clock enable
o_reset_ppg_start_up	output		reset ppg start up
o_ioff_calibration_av	output		ioff calibration data valid
o_mc_busy	output		main controller is working
o_half_charge_ldo	output		half charge ldo to pmu
o_start_su_ecg	output		start ecg start up
o_start_su_bia	output		start bia start up
o_alc_mask	output		alc mask
o_start_cfg_update_clk_en	output		start shadow registers IPs
o_save_alc	output		save alc data for current frame

Port name	Direction	Type	Descr.
o_irq_start_measurement	output		irq for start measurement
o_reset_su	output		reset start up phase
o_operative_state	output		operative state flag
o_ioff_rx_ch_reg	output		ioff rx ch selection during ppg calibration
o_rst_async_ecg_reg	output		start rst async reset ecg
o_rst_async_bia_reg	output		start rst async reset bia
o_write_data_cfg	output		writa data change ppg configuration
o_write_cfg_change_ppg	output		start write configuration change for ppg
o_su_ecg_clk_en	output		start up phase ecg clock enable

Port name	Direction	Type	Descr.
o_su_bia_clk_en	output		start up phase bia clock enable
o_elab_fsm_clk_en	output		elab fsm clock enable
o_rx_ch_gain_calib	output	[rx_ch_gain_max_bit-1:0]	o rx channel gain for ppg ioff calibration
o_en_mask_ldo_pdclkf_reg	output		mask ldo control during boot
o_boot_state_sync	output		boot_state sync on 16MHz
o_ispu_irq_ctrl	output	[3:0]	ispu irqs
o_cfg_enable	output		enabling shadow register
o_check_update	output		check update configuration
o_first_cfg	output		first configuration
o_start_operative_16MHz	output		start operative

Port name	Direction	Type	Descr.
			mask for Lead Off detection
o_main_controller_device_status_register	output	[2:0]	device fsm status register
o_main_controller_elab_status_register	output	[3:0]	elab fsm status register
o_en_bufxbio	output		en biobuffer to analog domain
o_write_time_slot_reg	output		write time slot data start operation
o_time_slot_data	output	[23:0]	time slot data
o_cfg_change_ppg_data	output	[23:0]	change ppg data
o_en_lead_off	output		enable lead off
o_start_ahb_time_slot_write	output		start ahb time slot ahb write
o_gsr_clk_en_16M	output		gsr clock enable 16 MHz

Port name	Direction	Type	Descr.
o_start_operation_for_ppg	output		start operation for ppg
o_ppg_ioff_calib_start	output		start ioff ppg calib
o_ppg_adc_control_en	output		ppg adc control
o_clock_en_boot_sync_32	output		clock enable boot sync
o_reset_alc	output		reset alc

Table 6: Main Controller FSMs I/O List

5.3.2. Description of the Block

This block is divided into two FSMs. The first FSM called Device_FSM, is in charge of identify the working state of the device. The second FSM called Elab_FSM, is in charge of identify what signal is in elaboration and to check and generate signal from and for the Analog Domain. The Elab_FSM is also in charge to start writing to the AHB Crossbar and to talk with Hardware DSPs block like the Avarage Circuit and ADSP, or to start PPG operation talking with the PPG_FSM. This Block is also in charge to turn of the Fast Ring Oscillator (16 MHz) and to enable the User Read Operation.

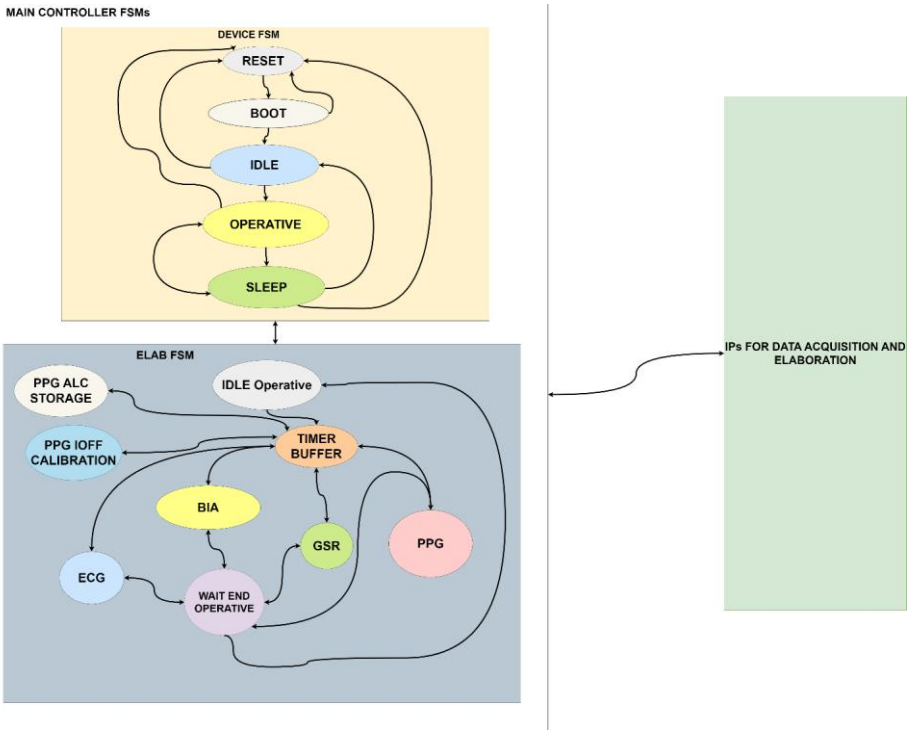


Figure 2: Main Controller FSM Diagram

5.3.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_003] ASSUMPTION:
When in BOOT State (when main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x01), the Device_FSM shall raise the w_quokka_boot signal let the ADSP starts the BOOT operation. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_5014] REQUIREMENT:
When the MODE OPERATION is set to 0 (after that has been set to 1) to stop a measurement, the USER shall check that the DEVICE_STATUS_REGISTER reaches the IDLE State Value and wait at least 1ms before doing anything. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_5015] Requirement:
[Covers: DDS_STBIO1_1114]
The State WAIT_SU shall not be considered interruptible. If the user wants to interrupt the WAIT_SU state, it shall perform the Soft Reset procedure described in the DDS [DDS_STBIO1_1114] [END]

[IPOS_STBIO1_MAIN_CONTROLLER_5016] REQUIREMENT:

When an ECG, BIA or GSR Operation is selected and DS_SEL is set to 0, The DATA Output for these two channels shall be written in the following regmap registers

ECG0_AC
ECG0_DC
ECG1_BIA_P_AC
ECG1_BIA_P_DC
ECG2_BIA_Q_AC
ECG2_BIA_Q_DC
GSR_RESULT

[END]

[IPOS_STBIO1_MAIN_CONTROLLER_5004] REQUIREMENT:
The max value that the user shall use for every m_channel (m_ecg, m_bia, m_gsr and m_ppg) is 1024[END]

[IPOS_STBIO1_MAIN_CONTROLLER_5005] REQUIREMENT:
If the user set a value greater than 1024 into ones of the m_channel (see req 5004), the value shall clamp at 1024 (check the input i_m_ecg, i_m_bia, i_m_gsr or i_m_ppg into the main controller fsm)[END]

[IPOS_STBIO1_MAIN_CONTROLLER_1004] REQUIREMENT:
[Covers: DDS_STBIO1_0118]
[Covers: DDS_STBIO1_0125]
The user shall configure time slot duration in DEVICE_CONFIG_OPERATION register [4:2] [END]
[IPOS_STBIO1_MAIN_CONTROLLER_1005] REQUIREMENT:
The user shall configure which channel is selected and if work in Data Storage mode for the measure writing the DEVICE_CONFIG_CHANNEL register [Covers] [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1006] REQUIREMENT:
The user shall select a division index for ECG, BIA, GSR and PPG configuring M_ECG_l, M_ECG_h, M_BIA_l, M_BIA_h, M_GSR_l, M_GSR_h, M_PPG_l and M_PPG_h writing in the dedicated registers. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1007] REQUIREMENT:
[Covers: DDS_STBIO1_0120]
[Covers: DDS_STBIO1_0122]
[Covers: DDS_STBIO1_0127]
[Covers: DDS_STBIO1_0130]
In Order to enable an ECG0 Channel Operation, the user shall always select the ECG0 channel (DEVICE_CONFIG_CHANNEL), set the i_M_ECG greater than zero and set to HC_EN. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1008] REQUIREMENT:

[Covers: DDS_STBIO1_0128]
[Covers: DDS_STBIO1_0140]
In Order to enable a BIA Channel Operation, the user shall always select the BIA channel (DEVICE_CONFIG_CHANNEL), set the i_M_BIA greater than zero and set to IMP_EN. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1009] REQUIREMENT:
In Order to enable a GSR Channel Operation, the user shall always select the GSR channel (DEVICE_CONFIG_CHANNEL) and set the i_M_GSR greater than zero. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1010] REQUIREMENT:
In Order to enable a PPG Channel Operation, the user shall always select the PPG channel (DEVICE_CONFIG_CHANNEL) and set the i_M_PPG greater than zero. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1011] REQUIREMENT:
In Order to enable an ECG1_ECG2 Channel Operation, the user shall always select the ECG1_ECG2 channel (DEVICE_CONFIG_CHANNEL), set the i_M_ECG greater than zero and IMP_EN set to 1. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_004] REQUIREMENT:
[Covers: DDS_STBIO1_0116]
When the Device_FSM FSM is in BOOT State, that is DEVICE_STATUS_REG register [2:0]) == 0x01 (main_controller_status_register), it shall wait that the ADSP ends the BOOT Phase, checking when the i_quokka_boot_end is set high. When the BOOT Phase is ended, the Device_FSM FSM shall go to IDLE State, check that DEVICE_STATUS_REG register [2:0]) == 0x02, (main_controller_status_register), check that the w_en_ldo1v8 signal is set low and that the O_PD_CLKF signal is set to high. [End]

[IPOS_STBIO1_MAIN_CONTROLLER_008] REQUIREMENT:
When in unique_fifo_mode (i_multiple_fifo set to 0), if during a time slot an operation is done, the Main_Controller_FSM shall write the time_slot_data inside the FIFO with a dedicated tag. The value of the time_slot_data shall be referred to the start of the entire measurement cycle (for example, time_slot_data shall be 0 if it's the first time slot, 4 if it's the fifth time_slot). If no operation is done during a time_slot, no time_slot_data shall be written, but the value of time_slot_data shall be increased. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_009] REQUIREMENT:
[Covers: DDS_STBIO1_0123]
[Covers: DDS_STBIO1_0141]

When in IDLE State, if ECG0_SEL (DEVICE_CONFIG_CHANNEL register [0]) is high, or BIA_SEL (DEVICE_CONFIG_CHANNEL register [2]) is set high, or ECG1_ECG2_SEL (DEVICE_CONFIG_CHANNEL register [1]) is set high, or GSR_SEL (DEVICE_CONFIG_CHANNEL register [3]) is set high, or ADC_DO_CALIBRATION (

ADC_CAL_CONFIG register [0]) is set high, or WAVE_VELOCITY_MODE (GENERAL_PPG_PARAMETERS_1 register [4]) is set high together with PPG_SEL (DEVICE_CONFIG_CHANNEL register [4]), when MODE_OPERATION (DEVICE_CONFIG_OPERATION register [1:0]) is set different from 0, the O_PD_CLKF shall go to 0, the w_en_ldo1v8 shall be set to 1. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_010] REQUIREMENT:
[Covers: DDS_STBIO1_0119]
When the FSM is in IDLE State, if ECG0_SEL (DEVICE_CONFIG_CHANNEL register [0]) is high, HC_EN (ECG_CTRL register [0]) is set to 1 and IMP_EN (IMP_CTRL register [0]) is set to 0, ECG0 shall be active so, when MODE_OPERATION (DEVICE_CONFIG_OPERATION register [1:0]) is set different from 0, the O_PD_HC2 shall go low and the Device_FSM FSM shall go to the WAIT_SU State, so the main_controller_status_register (DEVICE_STATUS_REG register [2:0]) shall be equal to 0x03. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_011] REQUIREMENT:

When the HC_EN (ECG_CTRL register [0]) is set to 0 and IMP_EN (IMP_CTRL register [0]) is set to 1 and ECG1_ECG2_SEL (DEVICE_CONFIG_CHANNEL register [1]) is set to 1, ECG1 and ECG2 shall be active so, when MODE_OPERATION (DEVICE_CONFIG_OPERATION register [1:0]) is set different from 0, the O_PD_HC2 shall remain high, O_PD_IMP shall go low and the Device_FSM shall go to the WAIT_SU State, so main_controller_status_register (DEVICE_STATUS_REG register [2:0]) shall be equal to 0x03. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_012] REQUIREMENT:
When the HC_EN (ECG_CTRL register [0]) is set to 1 and IMP_EN (IMP_CTRL register [0])is set to 1, ECG1_ECG2_SEL (DEVICE_CONFIG_CHANNEL register [1]) is set to 1 and ECG0_SEL (DEVICE_CONFIG_CHANNEL register [0]) is set to 1, ECG0-ECG1 and ECG2 shall be active so the O_PD_HC2 shall go low, O_PD_IMP shall go low and the Device_FSM shall go to the WAIT_SU State so main_controller_status_register (DEVICE_STATUS_REG register [2:0]) shall be equal to 0x03. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_015] REQUIREMENT:
When the user selects the ECG, BIA, the wave velocity mode or the ppg ioff calibration, the device Shall go to WAIT_SU state. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_0115] REQUIREMENT:

Commented [AI2]: @Aldo TORINO thisi is 1 requirement. You shall stop here, adding a detailed reference on how you can check the start-up sequence started. The rest shall go to a second Req_ID.

When the user selects an operative mode that includes the BIA channel, The DEVICE_FSM FSM moves from IDLE STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x02) to WAIT_SU STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x03). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_016] REQUIREMENT:
When DEVICE is in WAIT_SU, if IMP_EN (IMP_CTRL register [0]) is equal to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL [2]) is equal to 1, O_PD_IMP shall go low within 5 clock cycle at 32kHz. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_017] REQUIREMENT:
When DEVICE is in WAIT_SU, if IMP_EN (IMP_CTRL register [0]) is equal to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL register [2]) is equal to 1 and the IMP_CURRINJ_EN (CURR_INJ register [3]) is equal to 1, the O_IMP_CURINJ_PD shall go low. On the other hand, if IMP_CURRINJ_EN (CURR_INJ register [3]) is equal to 0, the O_IMP_CURINJ_PD shall remain high. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_018] REQUIREMENT:
When DEVICE is in WAIT_SU, if IMP_EN (IMP_CTRL register [0]) signal is equal to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL register [2]) is equal to 1, 5ms after the falling of imp_curinj_pd and O_PD_IMP, if IMP_CURRINJ_EN (CURR_INJ register [3]) is equal to 1, O_CK_M, O_CK_P and O_CK_Q shall be generated from Clock generator block so they run at FREQ_BIA_CLK_m_p_q (EN_CHOP_FREQ_MOD_DEMOD register [4:3]) and its values shall be chosen from 20kHz, 50kHz, 100kHz, 200kHz .
On the other hand, if IMP_CURRINJ_EN (CURR_INJ register [3]) is equal to 0, ck_m, ck_p and ck_q shall stay low. After this, the clock shall always run. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_019] REQUIREMENT:
When IMP_CURRINJ_EN (CURR_INJ register [3]) is high, O_CK_Q shall be run with a phase delay of 90 degrees (opposite) with respect O_CK_P. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_020] REQUIREMENT:
When DEVICE_FSM FSM is in WAIT_SU state, if IMP_EN (IMP_CTRL register [0]) is equal to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL register [2]) is equal to 1, 15ms after the falling of O_PD_IMP, the O_RST_IMP shall go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_021] REQUIREMENT:
When DEVICE_FSM FSM is in WAIT_SU state, if IMP_EN (IMP_CTRL register [0]) is equal to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL register [2]) is equal to 1; 20ms after the falling of O_RST_IMP, if EN_CHOP_CLK_IMP (EN_CHOP_FREQ_MOD_DEMOD register [1]) is set to 1,

Commented [AI3]: @Aldo TORINO can you please which register shall be set and how ?

Commented [AT4R3]: It's not a register, but it's moved from Main Controller.

Commented [AI5]: @Aldo TORINO immediately or in the next clock cycles ? Please clarify. The SVA change a lot.

Commented [AT6R5]: Modified

Commented [AI7]: @Aldo TORINO please clarify how to reach this functionality: which register to be set and how ?

Commented [AT8R7]: It's generated from Clock Gen in PMU.

O_CK_CHOP_IMP shall be generated from Clock generator block, so O_CK_CHOP_IMP shall be run at 10kHz frequency.
On the other hand, if EN_CHOP_CLK_IMP (EN_CHOP_FREQ_MOD_DEMOD register [1]) is set to 0, then O_CK_CHOP_IMP shall stay low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_022] REQUIREMENT:
When DEVICE_FSM FSM is in WAIT_SU state, if IMP_EN (IMP_CTRL register [0]) is equal to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL register [2]) is equal to 1; 30ms after the falling of O_RST_IMP, O_RSTI_IMP shall go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_023] REQUIREMENT:
When DEVICE_FSM FSM is in WAIT_SU state, if IMP_EN signal (IMP_CTRL register [0]) is equal to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL register [2]) is equal to 1; If i_su_hlt_dly is set to 0, 205ms after the falling of O_RSTI_IMP, O_SUG_IMP shall go low.
Instead, if SU_DLY (EN_CHOP_FREQ_MOD_DEMOD register [2]) is set to 1, 755ms after the falling of O_RSTI_IMP, O_SUG_IMP shall go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_024] REQUIREMENT:
When DEVICE_FSM FSM is in WAIT_SU STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x03) if HC_EN (ECG_CTRL register [0]) is set to 1 and ECG0_SEL (DEVICE_CONFIG_CHANNEL register [0]) is set to 1 and MODE_OPERATION (bit [1:0] on DEVICE_CONFIG_OPERATION register) is different to 0, after WAIT_SU state the device shall go in OPERATIVE (DEVICE_STATUS_REG register [2:0] is equal to 0x04)[END]
[IPOS_STBIO1_MAIN_CONTROLLER_3124] REQUIREMENT:

When the ELAB_FSM is in ECG State and only ECG0 is selected, the o_ADC_EN shall go high, 5 acquisitions are required (acquisitions described in section 5.4 ADC PHASE). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_025] REQUIREMENT:
When DEVICE_FSM FSM is in WAIT_SU STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x02) if IMP_EN (IMP_CTRL register [0]) is set to 1 and ECG1_ECG2_SEL (DEVICE_CONFIG_CHANNEL [1]) is set to 1, after WAIT_SU the device_fsm shall go in OPERATIVE (DEVICE_STATUS_REG register [2:0] is equal to 0x04). [END]

[IPOS_STBIO1_MAIN_CONTROLLER_3125] REQUIREMENT:
When ELAB_FSM is in ECG state and ECG1 and ECG2 are selected, o_ADC_EN shall go high and 10 acquisitions are required (acquisitions described in chapter in section 5.4 ADC PHASE). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_026] REQUIREMENT:

Commented [AI9]: @Aldo TORINO which sequence it is needed to be followed and checked ?

Commented [AT10R9]: Described in dedicated paragraph

Commented [AI11]: @Aldo TORINO which sequence it is needed to be followed and checked ?

Commented [AT12R11]: In the section described in the hyperlink

When DEVICE_FSM FSM is in IDLE STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x02) if HC_EN (ECG_CTRL register [0]) is set to 1 and IMP_EN (IMP_CTRL register [0]) is set to 1 and ECG0_SEL (DEVICE_CONFIG_CHANNEL [0]) is set to 1 and ECG1_ECG2_SEL (DEVICE_CONFIG_CHANNEL [1]) is set to 1, after WAIT_SU the device_fsm shall go in OPERATIVE (DEVICE_STATUS_REG register [2:0] is equal to 0x04).[END]

[IPOS_STBIO1_MAIN_CONTROLLER_3126] REQUIREMENT:
When ELAB_FSM is in ECG state and EC0, ECG1 and ECG2 are selected, o_ADC_EN shall go high and15 acquisitions are required (acquisitions described in 5.4 ADC PHASE) and then device shall go in SLEEP (DEVICE_STATUS_REG register [2:0] is equal to 0x05). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_027] REQUIREMENT:
When DEVICE_FSM FSM is in IDLE STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x02) if IMP_EN (IMP_CTRL register [0]) is set to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL [2]) is set to 1, after WAIT_SU the device_fsm shall go in OPERATIVE (DEVICE_STATUS_REG register [2:0] is equal to 0x04).[END]

[IPOS_STBIO1_MAIN_CONTROLLER_3127] REQUIREMENT:
When ELAB_FSM is in BIA state, o_ADC_EN shall go high and 16 acquisitions are required (acquisitions described in section 5.4 ADC PHASE) and then device shall go in SLEEP (DEVICE_STATUS_REG register [2:0] is equal to 0x05). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_028] REQUIREMENT:
When DEVICE_FSM FSM is in IDLE STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x02) if HC_EN (ECG_CTRL register [0]) is set to 1 and ECG0_SEL (DEVICE_CONFIG_CHANNEL [0]) is set to 1 and IMP_EN (IMP_CTRL register [0]) is set to 1 and BIA_SEL (DEVICE_CONFIG_CHANNEL [2]) is set to 1, after WAIT_SU, the device FSM shall go in OPERATIVE (DEVICE_STATUS_REG register [2:0] is equal to 0x04). [END]

[IPOS_STBIO1_MAIN_CONTROLLER_128] REQUIREMENT:
When ECG0 and BIA are selected 21 acquisitions shall be required (acquisitions described in section 5.4 ADC PHASE) and then device shall go in SLEEP (DEVICE_STATUS_REG register [2:0] is equal to 101). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_029] REQUIREMENT:
When DEVICE_FSM FSM is in IDLE if PPG_SEL (DEVICE_CONFIG_CHANNEL [4]) is set to 1 and MODE_OPERATION (bit [1:0] on DEVICE_CONFIG_OPERATION register) is different to 0, the device shall go in PPG state (main_controller_status_register in ELAB_STATUS_REG register [3:0] shall be equal to 0x05) and O_PD_TIA and O_PD_IDAC shall go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_030] REQUIREMENT:
When DEVICE_FSM FSM is in PPG state (main_controller_status_register in ELAB_STATUS_REG register [3:0] shall be equal to 0x05) the device shall be execute a number of frames equal to N_PPG_FRAME (GENERAL_PPG_PARAMETERS_1 register [3:0]) (Check the i_id_frame shall increase until it's equal to N_PPG_FRAME-1). At the beginning of each frame o_ADC_EN shall go high and it shall go low when ALC_COMPENSATION is performed or at the end of the last frame. **[END]**

Commented [AI13]: @Aldo TORINO which sequence it is needed to be followed and checked ?

Commented [AT14R13]: In the section described in the hyperlink

Commented [AI15]: @Aldo TORINO which sequence it is needed to be followed and checked ?

Commented [AT16R15]: In the section described in the hyperlink

Commented [AI17]: @Aldo TORINO which sequence it is needed to be followed and checked ?

Commented [AT18R17]: In the section described in the hyperlink

Commented [AI19]: @Aldo TORINO how we can check the Device is executing that number of frames ? Which registers or signals to be checked ?

Commented [AT20R19]: Fixed

[IPOS_STBIO1_MAIN_CONTROLLER_031] REQUIREMENT:
When DEVICE_FSM FSM is in PPG state (main_controller_status_register in ELAB_STATUS_REG register [3:0] shall be equal to 0x05) o_ADC_start shall go high a number of times equal "2^ value indicated in regmap of FRAME_{n}_SAMPLES (FRAME_n_PARAMETERS_1 register [5:3]) for ALC and PPG both (2*(2^regmap_value). Except when FRAME_{n}_SAMPLES is equal to 3'b000, in this case o_ADC_start shall go high 3 times. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_032] REQUIREMENT:
When DEVICE_FSM FSM is in PPG state (main_controller_status_register in ELAB_STATUS_REG register [3:0] shall be equal to 0x05) during each frame o_Txi_idac_VREF shall go from 0 to N_MAX (DIGITAL_RAMP_CONFIG_PARAM_1 register [3:0]). At first, o_Txi_idac_VREF shall be increase its value of N_START for next step o_Txi_idac_VREF shall be increase its value by 1 **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_033] REQUIREMENT:
When not in IDLE STATE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x02), Device_FSM shall go back in IDLE only if the user turns off the MODE_OPERATION (DEVICE_CONFIG_OPERATION register [1:0]) is set to 0). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_034] REQUIREMENT:
When in OPERATIVE (main_controller_status_register (DEVICE_STATUS_REG register [2:0]) is equal to 0x02), if the time slot counter reach its limit, the Device_FSM shall go into the error state (DEVICE_STATUS_REGISTER) and raise an interrupt (o_error_time_slot) to the Digital_Top. **[END]**

Commented [AI21]: @Aldo TORINO please specify which signal shall be set

Commented [AT22R21]: done

[IPOS_STBIO1_MAIN_CONTROLLER_4034] REQUIREMENT:
Every Time a new measurement cycle starts (DEVICE FSM goes in OPERATIVE from another state), the o_irq_start_measurement shall be set to 1 for at least 1 clock cycle **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1034] REQUIREMENT:
Once in the ERROR state, the user shall wait 1ms before set to 0 the MODE_OPERATION to go back into the IDLE State. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_035] REQUIREMENT:
When DEVICE_FSM FSM is in Operative State (DEVICE_STATUS_REG register [2:0]==0x04), the o_MC_busy shall go to 1 and shall go to 0 one clock cycle after entering the SLEEP state if no error occurred. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_036] REQUIREMENT:

When in Sleep Mode, the device shall go back to the Operative State if no error (o_error_irq stuck at 0) occurs and all the operations are ended. Furthermore, if no Bio-Channel operations are selected and more than 150us are left before the start of a new time slot, the o_pd_clkf (Digital Top signal) shall be set to 1, otherwise shall be left to 0. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_436] REQUIREMENT:
When in Sleep Mode and no Bio-Channel are selected, if every elaboration is ended and there are at least 34 clock cycles at 32kHz before the start of a new time slot, the o_en_ldo_1v8 shall go low. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_037] REQUIREMENT:
In a Time Slot, when a BIO-Channel operation is selected, o_en_bufxbio signal (in Digital Top) shall go high at least a configurable time (Device Config Operation [6:5]) before selecting the first bio-channel to measure with the o_ADC_mux_reg. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_038] REQUIREMENT:
When the ELAB_FSM is in ECG State, the ADC_Mux (concatenation of {o_adc_mux_sel, o_adc_mux_bio_channel} on digital top) output shall be set equal to 001 and after 4 acquisitions (described in section 5.4 ADC PHASE) shall be set to 010. If ECG1 and ECG2 are selected, the o_adc_mux_biochannel shall be set also to 011, 100, 101 and 110. In the ECG state. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_039] REQUIREMENT:
When ELAB FSM is in PPG state (main_controller_status_register in ELAB_STATUS_REG register [3:0] shall be equal to 0x05) or a WAIT_END_OPERATIVE State, o_en_biobuffer shall stay low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_040] REQUIREMENT:
When the ELAB_FSM is in BIA State (Check ELAB_STATUS_REG), the ADC_Mux (o_adc_mux_biochannel on digital top) output shall be set equal to 011, 100, 101 or 110. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_041] REQUIREMENT:
When the ELAB_FSM is in GSR State (Check ELAB_STATUS_REG), the ADC_Mux (o_adc_mux_biochannel on digital top) output shall be set equal to 111. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_3128] REQUIREMENT:
When the wave velocity mode is selected and the device FSM is in WAIT_SU, the Elaboration FSM shall go into the TIMER BUFFER State and then into the PPG ALC STORAGE [END]

[IPOS_STBIO1_MAIN_CONTROLLER_3129] REQUIREMENT:
When the ppg ioff calibration and the adc calibration are selected, the adc calibration shall be performed first.[END]

[IPOS_STBIO1_MAIN_CONTROLLER_3130] REQUIREMENT:
When the wave velocity mode and the ppg ioff calibration are selected, the wave velocity mode shall be performed first. [END]

Commented [AI23]: @Aldo TORINO please clarify which value of which register

Commented [AI24]: @Aldo TORINO please clarify which signal to be checked

Commented [AI25]: @Aldo TORINO please specify which signal shall be set

Commented [AT26R25]: done

Commented [AI27]: @Aldo TORINO please clarify which signal to be checked

Commented [AI28]: @Aldo TORINO please clarify which signal to be checked

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Commented [AI31]: @Aldo TORINO please clarify which signal to be checked

Commented [AI32]: @Aldo TORINO please clarify the DEVICE_STATUS_REG expected value

Commented [AI33]: @Aldo TORINO please clarify the DEVICE_STATUS_REG expected value

[IPOS_STBIO1_MAIN_CONTROLLER_3131] REQUIREMENT:
When the ppg ioff calibration is selected and device fsm is in WAIT_SU State, the Elab FSM shall go into the Timer Buffer state and then PPG IOFF CALIB state. [END]

5.4. ADC PHGEN

5.4.1. IP Description

This block is in charge of generating and checking the signal useful for ADC calibration and Data conversion. This block has also three Debug Mode: debug_calibration, debug_adc_error_cap_bypass and debug_adc_readout.

5.4.2. I/O List

5.4.2.1. Parameters

Generic name	Type	Value	Description
N_clk_max_bit	int	6	
data_in_size_bit	int	16	
n_average_max	int	128	
n_average_max_cal	int	128	
n_average_debug	int	2	

5.4.2.2. Ports

Port name	Direction	Type	Description
i_clk	input		16 MHz Clock
i_rstn	input		Reset
i_N_clk	input	[N_clk_max_bit-1:0]	Clock cycles for sampling phase

Port name	Direction	Type	Description
i_ADC_en_mc	input		adc enable from main controller
i_start_ADC_mc	input		Start adc from main controller
i_start_ADC_ppg	input		start adc from PPG
i_ADC_EOC	input		End Of Conversion from Analog ADC
i_en_ADC_debug	input		enable for adc debug phase
i_do_calibration	input		calibration enable
i_adc_en_mask	input		mask for adc enable
i_ppg_adc_control_en	input		adc enable from ppg
i_debug_adc_error_cap8_0_bypass	input		debug enable for debug capacitance
i_adc_error_cap8_debug	input	[6:0]	cap debug value
i_adc_error_cap7_debug	input	[6:0]	cap debug value
i_adc_error_cap6_debug	input	[6:0]	cap debug value
i_adc_error_cap5_debug	input	[6:0]	cap debug value
i_adc_error_cap4_debug	input	[6:0]	cap debug value

Port name	Direction	Type	Description
i_adc_error_cap3_debug	input	[6:0]	cap debug value
i_adc_error_cap2_debug	input	[6:0]	cap debug value
i_adc_error_cap1_debug	input	[6:0]	cap debug value
i_adc_error_cap0_debug	input	[6:0]	cap debug value
i_reset_debug_adc_data	input		reset data sampled during debug
i_start_adc_debug	input		start adc in debug mode
i_debug_adc_readout_bypass	input		enabling for adc redout bypass
i_debug_calibration	input		enabling debug calibration
i_data_ADC_out	input	[data_in_size_bit-1:0]	data from adc
i_reset_end_calibration	input		reset calibration phase
i_n_average	input	[clog2(n_average_max_cal):0]	number of acquisition
i_calibration_phase	input		start calibration phase
i_noise	input		number of acquisition for noise
o_adc_test_cap	output	[9:0]	value of adc test cap
o_adc_error_cap9	output	[7:0]	calibration result

Port name	Direction	Type	Description
o_adc_error_cap8	output	[6:0]	calibration result
o_adc_error_cap7	output	[6:0]	calibration result
o_adc_error_cap6	output	[6:0]	calibration result
o_adc_error_cap5	output	[6:0]	calibration result
o_adc_error_cap4	output	[6:0]	calibration result
o_adc_error_cap3	output	[6:0]	calibration result
o_adc_error_cap2	output	[6:0]	calibration result
o_adc_error_cap1	output	[6:0]	calibration result
o_adc_error_cap0	output	[6:0]	calibration result
o_ADC_clk_en	output		adc phase clock enable
o_ADC_en	output		adc enable to analog domain
o_ADC_start	output		adc start to analog domain
o_EOC	output		end of conversion to average block
o_data_out	output	[data_in_size_bit-1:0]	sampled data
o_do_average	output		start average operation

Port name	Direction	Type	Description
o_data_out_debug_adc	output	[15:0]	data out in debug mode
o_end_cal_phase	output		end of calibration phase
o_adc_ready	output		adc ready signal

Table 7: ADC Phgen I/O List

5.4.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_042] ASSUMPTION:
The representation of the sampled signal is 16 bit in Two Complements. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_043] REQUIREMENT:
The ADC_PHGEN shall work at 16 MHz clock frequency. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_044] REQUIREMENT:
When i_rstn is asserted low, all the output shall be set to 0, except for count_cap that shall be set to 9 and o_end_cal_phase that shall be set to high. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_045] REQUIREMENT:
The o_adc_ready shall be set to 1 when the ADC is getting ready (o_adc_en is set to 1 but not 160 clock cycles are passed), or i_calibration_phase is set to 1 or i_adc_start is set to 1. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_046] ASSUMPTION:
The capacitor under calibration shall be read by the o_adc_test_cap[9:0] signal, which shall be use the one hot encoding. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1047] REQUIREMENT:
When the elab_fsm is in PPG State, the **I_PPG_ADC_CONTROL_EN** Shall be 0, otherwise can be 0 or 1. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_047] REQUIREMENT:
When the calibration phase is ended, the o_end_cal_phase shall be set to 1 and shall stay high until the reset is not asserted or the i_reset_end_calibration is set to 1. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_048] REQUIREMENT:
When the i_reset_end_calibration, i_do_calibration and i_mode_operation are set to 1, the o_end_cal_phase shall be set to 0. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_049] REQUIREMENT:
When ECG, BIA or GSR Is sampled, i_ADC_en is equal to 1, the o_ADC_clk_en shall be set to 1. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_051] REQUIREMENT:
[Covers: DDS_STBIO1_0121]
[Covers: DDS_STBIO1_0129]
The Sampling Phase shall be configurable by the user with i_N_clk values from 0 up to 63. This signal indicates the number of clock that o_ADC_start shall stay high. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_052] REQUIREMENT:
When the i_ADC_en_mc and i_adc_en_mask is set to 0, all the output shall be set to 0, except for the o_end_cal_phases and o_ADC_clk_en. The o_end_cal_phases shall be set to 1 if i_do_calibration (bit 0 of ADC_CAL_CONFIG) is set to 0, otherwise shall be set to 0 until the end of the calibration phase. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_0152] REQUIREMENT:
If the calibration has been done, the ADC_error_caps values computed shall reset only if the POR signal is asserted or shall be overwritten if another calibration is performed. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_053] REQUIREMENT:
When the i_ADC_en_mc and i_adc_en_mask is set to 1, at least 160 clock cycles shall pass before the ADC is ready to work. The first i_ADC_EOC shall be ignored. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_054] REQUIREMENT:
When the ADC_phases fsm is ready and an acquisition is required, the o_ADC_start shall stay High for i_N_clk+1 cycles. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_055] REQUIREMENT:
After an acquisition is started, the ADC_Phases shall wait for the i_ADC_EOC sampling the new data on the falling edge of the clock. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_057] REQUIREMENT:
When i_calibration_phase is set to 1 and ADC is ready, the calibration phase shall start. In this case, the o_end_cal_phase shall be asserted when the calibration phase is ended. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_058] REQUIREMENT:
The calibration phase shall set the adc_error_cap8…0 to the correct value, following the calibration algorithm. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_059] REQUIREMENT:
If the i_noise signal is set to 1 when the DEVICE_FSM is into PPG State, the i_n_avarage shall be divided by 2. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_6059] REQUIREMENT:

I_noise shall be set to 0 during the ADC calibration. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_060] REQUIREMENT:
When the last acquisition is performed, after the last i_ADC_EOC arrived, the ADC_phases shall set to high the o_do_avarage signal for 1 clock cycle. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_061] REQUIREMENT:
When the i_data_out_adc is sampled, the 15 bit (MSB in little endian), shall be negated in order to convert the input in Two Complement. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_062] REQUIREMENT:
When i_debug_calibration is set (second bitfield of ADC_CAL_CONFIG), the number of samples used by the avarage block during the calibration shall be set to 2. (The purpose of this mode is to speed-up the AM-S simulation). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_063] REQUIREMENT:
When i_debug_adc_error_cap8_0_bypass is set to 1 (set to 1 en_debug_adc_error_cap8_0_bypass of ADC_GENERAL_DEBUG_0), the calibration result shall be bypassed in favor of the value written inside the registers ADC_ERROR_CAP_x_DEBUG **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_064] REQUIREMENT:
When the adc_readout_bypass mode is enabled by the 2 bitfield ADC_GENERAL_DEBUG_0 , the sampling phase of the ADC Phgen shall be bypassed in favor of regmap. In particular, the Sampling phase shall be controlled by the register EN_ADC_DEBUG and EN_ADC_START in ADC_GENERAL_DEBUG_0 and the sampled data shall be written in ADC_DATA_OUT_DEBUG_LOW and ADC_DATA_OUT_DEBUG_HIGH. and In this mode, before start a new sampling operation, the RESET_DEBUG_ADC DATA in ADC_GENERAL_DEBUG_0 shall be set to 1 and then to 0. **[END]**

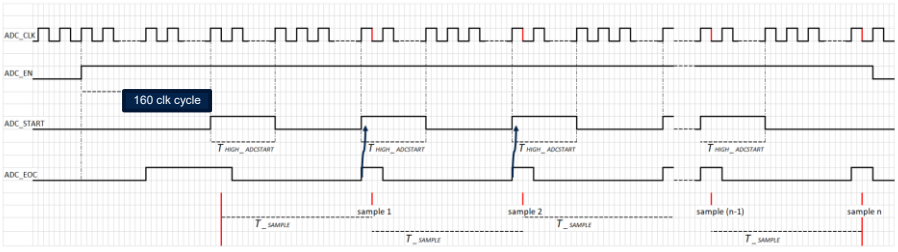


Figure 3: ADC PHGen Time Diagram:

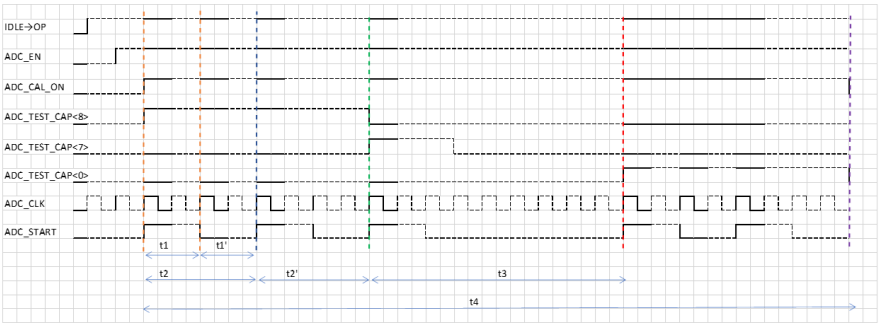


Figure 4: ADC Calibration Phase Time Diagram

$$ADC_{ERROR_{CAP9}} < 7:0 >= ADC_{DCAP9} < 15:8 >$$

$$ADC_{ERROR_{CAP8}} < 6:0 >= 0.5 \cdot (-ADC_{DCAP8} < 15:9 >) =$$

$$ADC_{ERROR_{CAP7}} < 6:0 >= 0.5 \cdot (-ADC_{DCAP7} < 15:9 > -ADC_{ERROR_{CAP8}} < 6:0 >) > \\ = -0.5 * ADC_{DCAP7} < 15:9 > +0.25 \cdot (ADC_{DCAP8} < 15:9 >)$$

$$ADC_{ERROR_{CAP6}} < 6:0 >= 0.5 \cdot (-ADC_{DCAP6} < 15:9 > -ADC_{ERROR_{CAP7}} < 6:0 > -ADC_{ERROR_{CAP8}} < 6:0 >) \\ = -0.5 * ADC_{DCAP6} < 15:9 > +0.25 * ADC_{DCAP7} < 15:9 > +0.125 \cdot (ADC_{DCAP8} < 15:9 >)$$

$$ADC_{ERROR_{CAP5}} < 6:0 >= 0.5 \cdot (-ADC_{DCAP5} < 15:9 > -ADC_{ERROR_{CAP6}} < 6:0 > -ADC_{ERROR_{CAP7}} < 6:0 > -ADC_{ERROR_{CAP8}} < 6:0 >) \\ = -0.5 * ADC_{DCAP5} < 15:9 > +0.25 * ADC_{DCAP6} < 15:9 > +0.125 \cdot (ADC_{DCAP7} < 15:9 >) + 0.0625 \cdot (ADC_{DCAP7} < 15:9 >)$$

……..

$$ADC_{ERROR_{CAP0}} < 6:0 >= 0.5 \cdot (-ADC_{DCAP0} < 15:9 > -ADC_{ERROR_{CAP1}} < 6:0 > -. \ldots . \ldots \ldots . -ADC_{ERROR_{CAP8}} < 6:0 >)$$

Figure 5: ADC Calibration Algorithm

[IPOS_STBIO1_MAIN_CONTROLLER_1064] ASSUMPTION:
ADC_CHANNEL TABLE:

Channel	o_adc_mux_bio_channel	o_adc_mux_sel
ECG0 AC	1	0
ECG0 DC	2	0
BIA AC P/ ECG1 AC	3	0
BIA DC P/ ECG1 DC	4	0
BIA AC Q/ ECG2 AC	5	0
BIA DC Q/ ECG2 DC	6	0
GSR	7	0
PPG	0	1
No Channel selection	0	0

Table 15: ADC Channel Table

[END]

5.5. ECG Start-Up Phase

5.5.1. IP Description

This block oversees the generation of the signal useful for the ECG start-up Analog Circuit

5.5.2. I/O List

5.5.2.1. Ports

Port name	Direction	Type	Description
i_clk	input		Clock general
i_rstn	input		reset general
i_hc_en	input		set health channel
i_clk_hlt_en	input		set clock for health channel
i_su_hlt_dly	input		start up phase delay
i_reg_rst_sync	input		reset async phase signal
o_pd_hc2	output		powerd down hc2 to analog domain
o_rst_hc2	output		rst hc2 to analog domain
o_rsti_hc2	output		rsti hc2 to analog domain
o_su_hc2	output		hc2 to analog domain
o_en_ck_hc2_10k	output		output clk chopper
o_reset_reg_rst	output		end of reset async phase

Table 88: ECG SU I/O List

5.5.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_072] REQUIREMENT:
When the i_hc_en is set to 1 and i_imp_en is set to 0 and i_n_ecg_channel is equal to 1, ECG0 shall be active so the o_pd_hc2 shall go low and counter shall be work. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_073] REQUIREMENT:
When the i_hc_en is set to 0 and i_imp_en is set to 1 and i_n_ecg_channel is equal to 2, ECG1 and ECG2 shall be active so the o_pd_hc2 shall remain high, o_pd_imp_from_bia shall go low and counter shall be work. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_074] REQUIREMENT:
When the i_hc_en is set to 1 and i_imp_en is set to 1 and i_n_ecg_channel is equal to 3, ECG0-ECG1 and ECG2 shall be active so the o_pd_hc2 shall go low, o_pd_imp shall go low and counter shall be work. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_075] REQUIREMENT:
The counter that defines the timing during the start-up phase shall work at 32KHz. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_076] REQUIREMENT:
15ms after that the o_pd_hc2 is gone low, the o_rst_hc2 shall go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_077] REQUIREMENT:
20ms after that the o_rst_hc2 is gone low, if i_clk_hlt_en is equal to 1, chopper_clock shall be generated and after another 10ms, o_rsti_hc2 shall go low. Instead, if i_clk_hlt_en is equal to 0, then o_rsti_hc2 shall go low after 30ms since the falling of o_rst_hc2. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_078] REQUIREMENT:
If i_su_hlt_dly is set to 0. At least after *205ms that the o_rsti_hc2 is gone low, o_su_hc2 shall go low.* **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_079] REQUIREMENT:
If i_su_hlt_dly is set to 1, at least after 755ms that the o_rsti_hc2 is gone low, o_su_hc2 shall go low .[END]
[IPOS_STBIO1_MAIN_CONTROLLER_4079] REQUIREMENT:
If i_rst_async_ecg is set to 1, the start-up phase shall restart but no clock shall stop if enabled. [END]
[IPOS_STBIO1_MAIN_CONTROLLER_4080] REQUIREMENT:
If i_rst_async_ecg is set to 1, it shall go to zero at least 5ms after that o_su_hc2 is set to 0. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_5080] REQUIREMENT:
i_rst_async_ecg Shall be used only when the Device FSM is in SLEEP or Operative State. [END]


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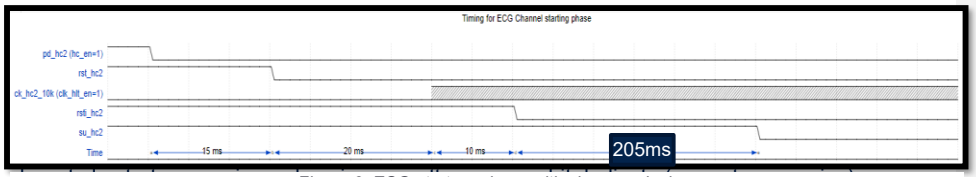


Figure 6: ECG start up phase with chopper clock

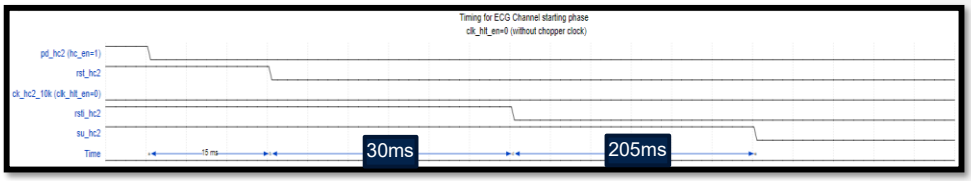


Figure 8:ECG Start Up phase with no Chopper Clock

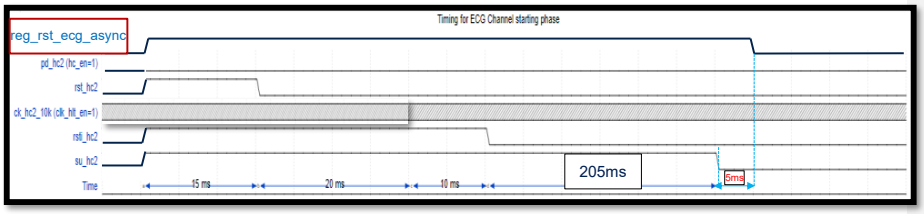


Figure 7:ECG Start-UP phase with asynch reset

5.6. BIA Start-Up Phase

5.6.1. IP Description

This block oversees the generation of the signal useful for the BIA start-up Analog Circuit

5.6.2. I/O List

5.6.2.1. Ports

Port name	Direction	Type	Description
i_clk	input		Clock general
i_rstn	input		reset general
i_imp_en	input		BIA En channel
i_imp_curinj	input		current injection for BIA
i_su_hlt_dly	input		delay for start up phase
i_clk_imp_en	input		clock BIA filter enable
i_reg_rst_imp_sync	input		reset sync from main controller fsm
o_pd_imp_from_bia	output		power down for analog domain
o_en_ck_m_p_q	output		clock for modulation
o_imp_curinj_pd	output		current injection into analog domain
o_rst_imp	output		reset bia for analog domain
o_en_ck_chop_imp	output		enable clock chopper for bia
o_rsti_imp	output		rsti for analog domain

Port name	Direction	Type	Description
o_sug_imp	output		sug for analog domain
o_reset_reg_rst_imp_async	output		end of reset async phase

Table 9: BIA Start-Up Phase I/O List

5.6.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_080] REQUIREMENT:
When the user selects an operative mode that includes the BIA channel, a start-up shall be done. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_081] REQUIREMENT:
When i_imp_en signal is equal to 1, the o_pd_imp_from_bia shall be go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_082] REQUIREMENT:
When i_imp_curinj signal is equal to 1, the o_imp_curinj_pd shall go low. On the other hand, if i_imp_curinj is equal to 0, the o_imp_curinj_pd shall stay high. **[END]**

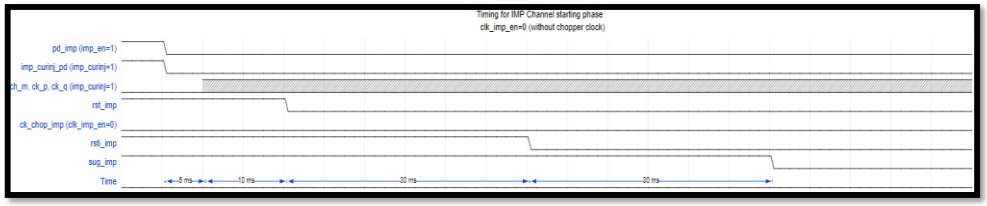
[IPOS_STBIO1_MAIN_CONTROLLER_083] REQUIREMENT:
At least 5ms after the falling of o_imp_curinj_pd and o_pd_imp_from_bia, if i_imp_curinj is equal to 1, o_en_ck_m_p_q shall be high and ck_m, ck_p and ck_q shall be generated from Clock generator block. On the other hand, if i_imp_curinj is equal to 0, the ck_m_p_q shall stay low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_084] REQUIREMENT:
At least 15ms after the falling of o_pd_imp_from_bia, the o_rst_imp shall go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_085] REQUIREMENT:
20ms after the falling of o_rst_imp, if i_clk_imp_en is set to 1, the o_en_ck_chop_imp shall be high and ck_chop_imp shall be generated from Clock generator block. On the hand, if i_clk_imp_en is set to 0, the o_en_ck_chop_imp shall stay low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_086] REQUIREMENT:
30ms after the falling of o_rst_imp, o_rsti_imp shall go low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_087] REQUIREMENT:
If i_su_hlt_dly is set to 0,at least 205 ms after the falling of o_rsti_imp, o_sug_imp shall go low. Instead, if i_su_hlt_dly is set to 1, 755ms after the falling of o_rsti_imp, o_sug_imp shall go low. **[END]**



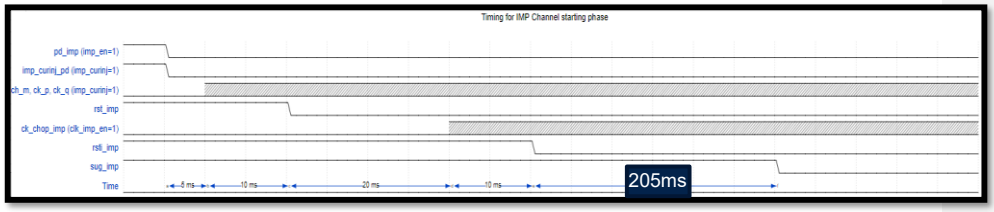
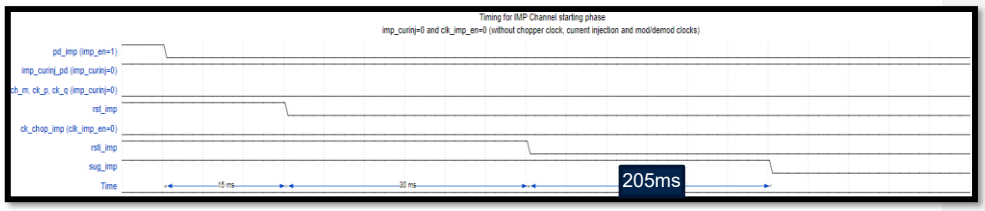
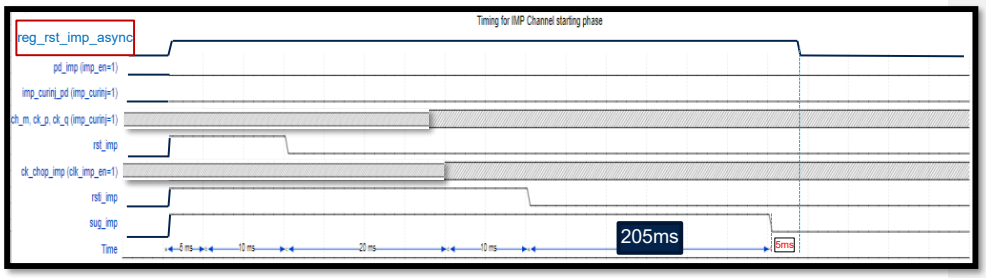


Figure 9: BIA Start-UP Phase Time Diagram



[IPOS_STBIO1_MAIN_CONTROLLER_4090] REQUIREMENT:
If i_rst_async_bia is set to 1, the start-up phase shall restart but no clock shall stop if enabled.[END]
[IPOS_STBIO1_MAIN_CONTROLLER_4091] REQUIREMENT:
If i_rst_async_bia is set to 1, it shall go to zero at least 5ms after that the o_sug_imp is set to 0 as in the next figure.[END]
[IPOS_STBIO1_MAIN_CONTROLLER_5091] REQUIREMENT:
i_rst_async_bia shall be used only when Device FSM is in OPERATIVE or SLEEP state.[END]

[IPOS_STBIO1_MAIN_CONTROLLER_5092] REQUIREMENT:
i_rst_async_bia shall be used also to reset the START-UP phase for ECG channel 1 and 2.[END]



5.7. GSR

5.7.1. IP Description

This is not an hardware block, but just a description of the GSR signal Behavior.

5.7.2. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_088] REQUIREMENT:
The i_m_gsr [SET_GSR_FREQ_h *and* SET_GSR_FREQ_l*] shall define how many time slots pass between two GSR acquisitions following this formula:*

$$\mathrm{GSR}_{\mathrm{data}} = \frac{\mathrm{time_slot}}{2 * \mathrm{m_gsr}}$$

.**[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_089] REQUIREMENT:
If the en_gsr_cds [bit 3 of GSR_CTRL] is set High, the i_m_gsr shall define also in which time slot after the GSR_Curr_on falls from 1 to 0, an acquisition without current injection (o_gsr_curr_on set to 0) shall be performed. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_090] REQUIREMENT:
The gsr_on shall set high at least 300us before the start of the sampling. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_092] REQUIREMENT:
If en_gsr_cds is set low [bit 3 of GSR_CTRL], the gsr_curr_on shall set high in every time slot where the GSR_SNS signal is acquired and shall set low at end of the same time slot. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_093] REQUIREMENT:
If en_gsr_cds is set high [bit 3 of GSR_CTRL] *the GSR_SNS (acquisition with o_gsr_curr_on set to 1) and GSR_CDS (acquisition with o_gsr_curr_on set to 0) shall be sampled 16 times, then mediated to get one single value for each of them, and then the final result shall be the subtraction between GSR_SNS and GSR_CDS*. On the other hand, If en_gsr_cds is set low, no acquisition of GSR_CDS shall be done and the result shall be the average of 16 acquisitions of GSR_SNS. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_094] REQUIREMENT:
When only GSR is selected (NO ECG0 and PPG are selected), if i_m_gsr [SET_GSR_FREQ_h *and* SET_GSR_FREQ_l*] is greater than 0, the Device FSM shall go in SLEEP after the IDLE State and shall go back in Operative State only when the measurement shall be run or, in case of cds disabled, when the o_gsr_curr shall change its polarity.* **[END]**

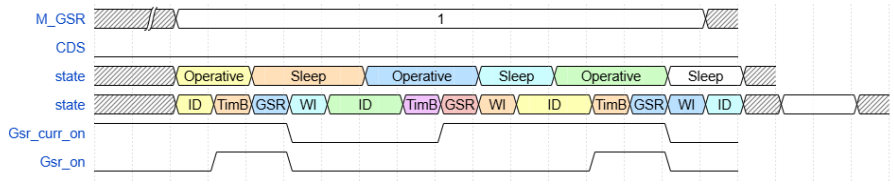


Figure 10: GSR with CDS Disabled and m_GSR set to 1

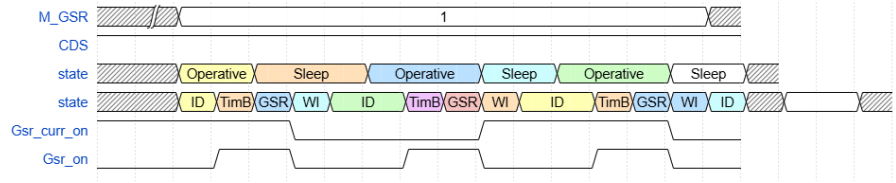


Figure 11:GSR with CDS Enabled with m_gsr set to 1

5.8. PPG FSM

5.8.1. IP Description

This IP is in charge to control the PPG TX/RX Enabling and signal acquisitions, and managing the start of Digital Ramp, ALC Compensation and PPG IOFF Calibration. Example of PPG IOFF Calibration Results can be found in the Appendix.

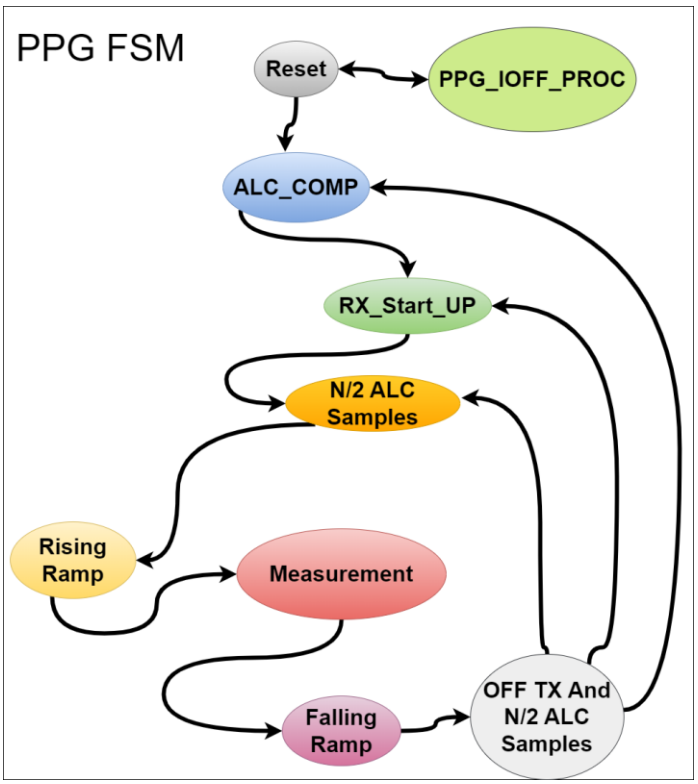


Figure 12: PPG FSM State Diagram

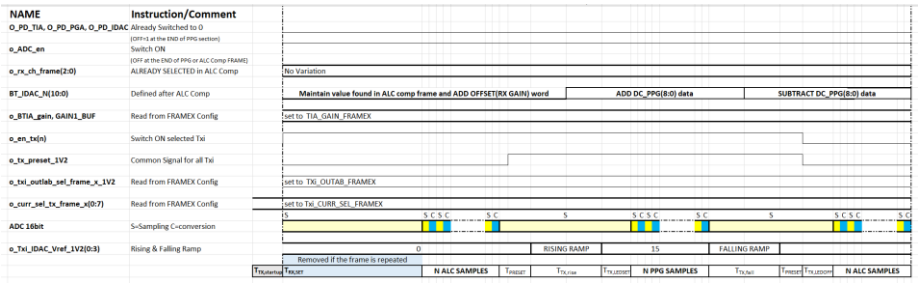


Figure 13: PPG Time Diagram

5.8.2. I/O List

5.8.3. Parameters

Generic name	Type	Value	Description
T_tx_startup_bit	int	5	
idac_dc_ppg_frame_bit	int	7	
rx_ch_frame_n_bit	int	6	
id_frame_max_bit	int	4	
rx_ch_gain_max_bit	int	4	
T_tx_led_set_bit	int	5	
T_tx_led_off_max_bit	int	7	
Nmax_bit	int	4	
ppg_ioff_offset_bit	int	10	
max_frame_repetition	int	2	
n_max_frame	int	12	
data_size_bit	int	16	
ppg_frame_bit	int	4	
T1_max_bit	int	9	
T1_max_bit_from_decoder	int	4	

Generic name	Type	Value	Description
BTIA_ALC_register_bit	int	10	
ppg_start_up_decod_bit	int	5	
ppg_start_up_bit	int	9	

5.8.4. Ports

Port name	Direction	Type	Description
i_clk_16MHz	input		Clk 16 MHz
i_rstn	input		Reset
i_gain_start_set	input	[rx_ch_gain_max_bit-1:0]	gain max value
i_T_rx_set	input	[T1_max_bit_from_decoder-1:0]	RX set Time
i_idac_dc_ppg_frame	input	[idac_dc_ppg_frame_bit-1:0]	IDAC DC PPG Value From Regmap
i_wave_velocity_mode	input		wave velocity mode selection
i_n_ppg_frame	input	[ppg_frame_bit-1:0]	number of ppg frame
i_const_value_ioff	input	[data_size_bit-1:0]	constant value for
i_adc_ready	input		adc ready from adc phases
i_start_ppg_fsm	input		start ppg fsm
i_reset_alc	input		reset alc compensation

Port name	Direction	Type	Description
i_start_operation_for_ppg	input		start operation for ppg
i_ppg_start_up_time	input	[ppg_start_up_decod_bit-1:0]	ppg start up time
i_rx_ch_frame	input	[rx_ch_frame_n_bit-1:0]	rx channel selection
i_id_frame	input	[id_frame_max_bit-1:0]	frame identification
i_T_tx_led_set	input	[T_tx_led_set_bit-1:0]	tx led set time
i_T_tx_led_off	input	[T_tx_led_off_max_bit-1:0]	tx led off time
i_reset_ppg_ioff_calib	input		reset ppg ioff calibration
i_reset_ppg_start_up	input		reset ppg start up phase
i_n_m0	input	[data_size_bit-1:0]	nm0 value
i_n_m1	input	[data_size_bit-1:0]	nm1 value
i_ppg_ioff_calibration	input		ppg ioff calibration mode
i_end_average	input		end average
i_rising_ramp_end	input		end of rising ramp
i_falling_ramp_end	input		end of falling ramp
i_half_rising_ramp	input		half value of the max digital ramp reached
i_end_compensation	input		end alc

Port name	Direction	Type	Description
i_done_frame	input		end of the frame
i_w_BTIA_ALC	input	[BTIA_ALC_register_bit-1:0]	ALC Coarse value
i_frame_repetition	input	[n_max_frame-1:0]	number of frame repetition
i_ppg_ioff_calib_mode	input		
o_data_valid_ppg_adc	output		data valid for ppg raw
o_data_valid_noise_ppg_adc	output		data valid for noise ppg
o_start_alc_comp	output		start alc comp
o_start_average_clk	output		enable average clock
o_start_digital_ramp	output		start rise digital ramp
o_falling_digital_ramp	output		start falling ramp
o_end_ppg_fsm	output		end of ppg fsm
o_en_tia	output		enable tia
o_en_buffer	output		enable buffer
o_en_idac	output		en idac
o_idac_dc_ppg_frame	output	[idac_dc_ppg_frame_bit-1:0]	idac dc ppg for current frame
o_RESET_PPG_average	output		reset ppg average

Port name	Direction	Type	Description
o_start_ADC	output		start ADC
o_reset_ramp	output		reset ramp value
o_busy_ppg	output		busy ppg signal
o_ppg_calib_data_n_m1	output		ppg calib data valid for nm1
o_alc_clk_en	output		alc compensation clock enable
o_tx_preset	output		set tx preset
o_en_txi	output		enable tx channel
o_average_repetion	output	[max_frame_repetition-1:0]	average repetition value
o_ppg_status_register	output	[n_state_bit-1:0]	ppg state status register
o_noise	output		ppg noise data valid
o_BTIA_ALC	output	[BTIA_ALC_register_bit-1:0]	btia alc value
o_reset_curr_frame	output		reset current frame
o_start_up_time_ppg_done	output		end of start up ppg
o_N_step2_reg	output	[ppg_ioff_offset_bit-1:0]	value stored for n_step2
o_mask_ADC_en	output		mask adc enable
o_en_rx_ch	output		enable rx channel

Port name	Direction	Type	Description
o_end_ioff_calibration	output		end ppg ioff calibration
o_ioff_offset	output	[ppg_ioff_offset_bit-1:0]	ioff offset value
o_ppg_calib_data_n_m0	output		ppg galibration data valid
o_reset_avg_calib_0	output		reset avg calib 0
o_reset_avg_calib_1	output		reset avg calib 1

Table 10: PPG FSM I/O List

5.8.5. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_7095] REQUIREMENT:
All the PPG_FSM state shall be checked using the PPG_STATUS_REGISTER in Regmap (the states values are described in the register's comment). **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_095] REQUIREMENT:
The o_En_TIA, o_EN_Buff, o_EN_ADC and o_EN_IDAC shall be high when the PPG_FSM is RX_START UP state. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_300] REQUIREMENT:
When the rstn_i is asserted low, the PPG FSM shall return in Reset state immediately. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_096] REQUIREMENT:
*When the i_m_ppg (*SET_PPG_FREQ_h *and* SET_PPG_FREQ_l*) is different from 0 and* PPG_SEL in DEVICE_CONFIG_CHANNEL register *is set to 1, when the ELAB_FSM is in PPG STATE, PPG_IOFF_CALIB or PPG_ALC_STORAGE, the* i_start_operation_for_ppg *shall be set to 1.* **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_097] REQUIREMENT:
When in Reset state, if the rstn_i is de-asserted , i_wave_velocity_mode is and the i_start_ppg_fsm is high, if i_ID_Frame is equal to 0 or two successively frame have two different i_rx_ch_frame, then the PPG FSM shall go to ALC Compensation state. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_6098] REQUIREMENT:
When i_wave_velocity_mode is set to 1, the ALC Compensation shall be set only when the ELAB FSM is into the PPG_ALC_STORAGE and then never repeated. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_6070] REQUIREMENT:
When i_wave_velocity mode is enabled together with the DISABLE_ALC, the PPG_ALC_STORAGE shall be performed but once the Device goes into OPERATIVE State, the values are not summed (visible) to the o_BIDAC. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_098] REQUIREMENT:
When in ALC Compensation, the PPG FSM shall raise the o_start_alc_comp *signal to the ALC Compensation block to start the ALC Compensation algorithm.* **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_099] REQUIREMENT:
When in ALC Compensation, the PPG shall wait that the rise of i_end_compensation signal from the ALC Compensation block to go into the RX_Start_UP state. This shall happen in OPERATIVE. **[END]**

IPOS_STBIO1_MAIN_CONTROLLER_6099] REQUIREMENT:
When in ALC Compensation, the PPG shall wait that the rise of i_end_compensation signal from the ALC Compensation block to go into the RESET_PPG state if i_wave_velocity_mode is set to 1. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_100] REQUIREMENT:
When in Rx_Start_Up state, the o_tx_Preset shall be set 0, the en_tx_i shall be set to 1, the o_BIDAC shall maintain the value reached during the ALC Compensation summed up, if i_ppg_ioff_calibration is enabled, with the IOFF_OFFSET computed during the IOFF Calibration Procedure. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_101] REQUIREMENT:
When in Rx_Start_Up state, the PPG_FSM shall stay in this state for the T_rx_set time (GENERAL_PPG_PARAMETERS_3 register [3:0])) and then go into the FIRST_ALC_SAMPLES state. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_102] REQUIREMENT:
When in FIRST_ALC_SAMPLES state, the o_start_ADC signal shall be set high until i_end_average is low. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_103] REQUIREMENT:
When in FIRST_ALC_SAMPLES *state, the PPG_FSM Shall stay in this state until 1 clock cycle after the rise of the* i_end_average and the TX_start_up time has passed*. After that, the PPG FSM shall bring itself into the Rising_Ramp state.* **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1102] REQUIREMENT:
Tx Preset shall be set to 1 a configurable TX_Preset_Step time before the start of the Digital Ramp rising phase[END]

[IPOS_STBIO1_MAIN_CONTROLLER_1103] REQUIREMENT:

Tx Preset shall be set to 0 0.5us after the Digital Ramp reach the 0 value during the falling phase [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1104] REQUIREMENT:
Tx Preset shall be set to 0 before the rising phase(i_start_digtal_ramp) and after the falling phase (i_start_falling_ramp) of the digital ramp [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1105] REQUIREMENT:
If no multiple frame is selected (i_frame_repetition in ppg_fsm set to 0), when Tx Preset falls from 1 to 0, also the en_tx shall pass from 1 to 0.[END]

[IPOS_STBIO1_MAIN_CONTROLLER_8106] REQUIREMENT:
If multiple frame is selected (i_frame_repetition in ppg_fsm greater than 0), when Tx Preset falls from 1 to 0, the en_tx shall pass from 1 to 0 only during the last repetition (check that when o_end_ppg_fsm is set, en_tx is 0).[END]

[IPOS_STBIO1_MAIN_CONTROLLER_104] REQUIREMENT:
When in FIRST_ALC_SAMPLES state, when i_end_average is set to 1, the o_start_digital_ramp signal shall be set high to start the rising of the digital ramp and go into the RISING_RAMP state. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_105] REQUIREMENT:
When in Rising_Ramp state, at the half of Rising_time, the o_idac_dc_ppg_frame shall be set to the IDAC_DC_PPG_FRAME_{n} (FRAME_n_PARAMETERS_5 register [5:0]) value related to the frame that is being executed. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_106] REQUIREMENT:
When in Rising_Ramp State, the PPG_FSM shall stay in this state until the end of the Rising Time and then go into the Measurement state. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_107] REQUIREMENT:
When in Measurement state, after the wait at least of the TX_ledset time related to the frame that is being executed. *(TX_LEDSET_FRAME {n}]), the o_start_adc signal shall be set high. [END]*

[IPOS_STBIO1_MAIN_CONTROLLER_108] REQUIREMENT:
When in Measurement state, the PPG_FSM shall stay in this state until the rising of the i_end_average signal from the ADC block. After that, the PPG_FSM shall bring itself into the Falling Ramp state. *[END]*

[IPOS_STBIO1_MAIN_CONTROLLER_109] REQUIREMENT:
When in MEASUREMENT State, the o_falling_digital_ramp signal shall be set to 1 to go into the FALLING_RAMP State and shall stay 1 until i_falling_ramp_end is different from 1. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_110] REQUIREMENT:
When in Falling_Ramp State, at the half of the Falling_Time the o_idac_dc_ppg_frame *shall be set to 0. [END]*

[IPOS_STBIO1_MAIN_CONTROLLER_111] REQUIREMENT:
When in Falling_Ramp state, the PPG_FSM shall stay in this state until the end of the Falling Time (depends on the user's configuration of Digital Ramp parameters) and bring itself into the SECOND_ALC_SAMPLES state. *[END]*

[IPOS_STBIO1_MAIN_CONTROLLER_112] REQUIREMENT:
When in SECOND_ALC_SAMPLES state, the o_en_Txi shall be set low after the end at least of the T_tx_ledoff time *(TX_LEDOFF_FRAME_{n})* and the o_start_adc signal shall be set to 1. *[END]*

[IPOS_STBIO1_MAIN_CONTROLLER_113] REQUIREMENT:
When in SECOND_ALC_SAMPLES state, the PPG_FSM shall stay in this state until the rise of the end_of_conversion signal from the ADC block. After that, if there are no more frames to run, the PPG_FSM shall go into the Reset State, otherwise, if there are other frames to run and channel rx is not changed, the PPG_FSM shall go into the Rx_start_up phase. On the other hand, if there are other frame to run and rx_channel is changed, the PPG_FSM shall go into the RESET_PPG state and the into the ALC_Compensation state. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1095] REQUIREMENT:
The o_pd_tia, o_pd_idac and o_pd_pga shall be the negated version the o_en_tia, o_en_idac and o_en_buffer of respectively. *[END]*

[IPOS_STBIO1_MAIN_CONTROLLER_115] REQUIREMENT:
At the end of each Frame, the start_avarage signal shall be rise to the Average block computation. Once the Average is computed, if no averages_between_equal_frames option is selected, the data shall be written inside the FIFO Memory using the Main Controller AHB Master interface. If there is avareges_between_equal_frames option selected from regmap, after the computations of this average, the obtained data shall be written inside the FIFO Memory using the Main Controller AHB Master interface. *[END]*

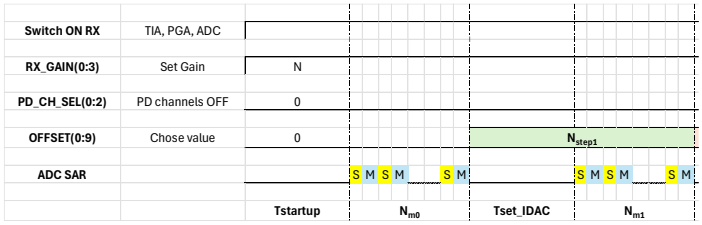


Figure 14: Ioff Calibration phase

$$N_{step2} = \left\lfloor \frac{N_{step1} * (N_{m0} - 2731)}{(N_{m0} - N_{m1})} \right\rfloor$$

[IPOS_STBIO1_MAIN_CONTROLLER_1115] REQUIREMENT

The Ioff Calibration algorithm shall be enabled or disabled by a dedicated field in regmap. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1116] REQUIREMENT:
When the Ioff Calibration algorithm is enabled, it shall be performed every time a new measurements cycle starts. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1118] REQUIREMENT:
Nm0 shall be the result of average on 8 acquired samples. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1119] REQUIREMENT:
Nm1 shall be the result of average on 8 acquired samples. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1120] ASSUMPTION
Nstep1 shall change as the following table:

RX Gain [ohm]	*Ideal OFFSET [uA]*	*MAX OFFSET [uA]*	*Typical Nstep2*	*Max Nstep2*	*Nstep1*
8000	*1.38E-04*	*1.65E-04*	*275*	*330*	*55*
16000	*6.88E-05*	*8.25E-05*	*137*	*165*	*27*
32000	*3.44E-05*	*4.13E-05*	*68*	*82*	*13*
64000	*1.72E-05*	*2.06E-05*	*34*	*41*	*7*
128000	*8.59E-06*	*1.03E-05*	*17*	*20*	*3*
256000	*4.30E-06*	*5.16E-06*	*8*	*10*	*1*
512000	*2.15E-06*	*2.58E-06*	*4*	*5*	*1*
1024000	*1.07E-06*	*1.29E-06*	*2*	*2*	*1*
2048000	*5.37E-07*	*6.45E-07*	*1*	*1*	*1*

Table 10: Nstep1 values

[IPOS_STBIO1_MAIN_CONTROLLER_1121] REQUIREMENT
During the Ioff calibration algorithm EN_TIA, EN_PGA and ADC_EN shall be set to 1. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1122] REQUIREMENT
During the Ioff calibration algorithm the RX_GAIN (o_BTIA_gain on top) shall change from 0 to 8. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1322] ASSUMPTION
Nstep2 is the result of the ioff calibration**[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1123] REQUIREMENT
During the Ioff calibration algorithm the PD_CH_SEL (o_rx_ch_frame on top) Shall be set to 0. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1124] REQUIREMENT
During the Ioff calibration algorithm the OFFSET (o_ioff_offset on ppg_fsm) shall change to 0 during the Nm0 computation and to Nstep1 during the Nm1 computation. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1125] REQUIREMENT
The Ioff Calibration algorithm shall be repeated 9 times, one for each RX gain value. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1126] REQUIREMENT
All the OFFSET Value shall be stored (See Ioff calibration Read Only registers in regmap) and used during the PPG acquisition based on the RX_GAIN configuration chosen by the user. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1127] ASSUMPTION
During the Ioff Calibration algorithm, all the PPG data (Nmo, Nm1, Nstep1 and final offset result) shall be considered unsigned. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_1128] ASSUMPTION
Outside the Ioff Calibration algorithm, all the PPG data shall be considered signed in two's complement. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1129] REQUIREMENT
When o_reset_state is set to 1, the o_BIDAC, o_BTIA_gain and o_rx_ch_frame (all checked from main_controller_top) shall be set to 0. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1130] ASSUMPTION
When o_reset_state falls from 1 to 0, the o_BIDAC, o_BTIA_gain and o_rx_ch_frame shall be considered stable. **[END]**
[IPOS_STBIO1_MAIN_CONTROLLER_2130] ASSUMPTION
The result of the IOFF CALIBRATION Nstep2 for rx gain equal to 2048000 shall be always 1. **[END]**

[IPOS_STBIO1_MAIN_CONTROLLER_1131] REQUIREMENT
To reset the IOFF_CALIBRATION PROCEDURE and relaunch it after that the mode Operation has been set to 0 and then again to 1, the user shall perform the following steps:

- Set the RESET_IOFF_CALIBRATION TO 1 (GENERAL_PARAMETERS1) to 1
- WAIT that the DEVICE_STATUS_REGISTER is equal to the OPERATIVE value and the ELAB_STATUS_REGISTER is equal to the PPG value
- Reset the RESET_IOFF_CALIBRATION FROM 1 (GENERAL_PARAMETERS1) to 0
- Set the MODE_OPERATION to 0
- Wait at least 1ms and the SET MODE_OPERATION to 1

[END]

5.9. Digital Ramp

5.9.1. IP Description

This block Is in charge to generate a user-configurable ramp for the PPG Analog Circuit

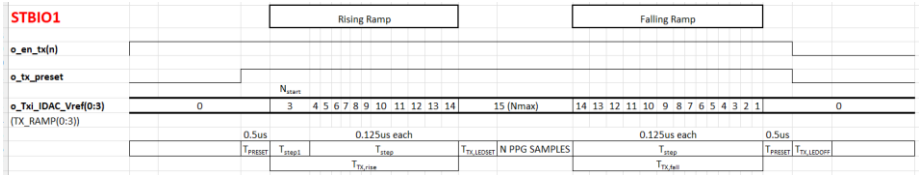


Figure 15: Digital Ramp Time Diagram

5.9.2. I/O List

5.9.2.1. Parameters

Generic name	Type	Value	Description
Nmax_bit	int	4	
delta_bit	int	2	

5.9.2.2. Ports

Port name	Direction	Type	Description
i_clk	input		Clock
i_rstn	input		Asynchronous reset active low
i_start_digital_ramp	input		start rising digital ramp
i_falling_digital_ramp	input		start falling digital ramp
i_Nmax_to_reg	input	[Nmax_bit-1:0]	

Port name	Direction	Type	Description
i_T_preset	input	[1:0]	Preset time
i_n_start_case	input	[Nmax_bit-1:0]	first step value for ramp
i_T_step1_to_reg	input	[delta_bit-1:0]	time for first step value for ramp
i_reset_ramp	input		reset ramp
o_rising_ramp_end	output		end of rising phase
o_falling_ramp_end	output		end of falling phase
o_Tx_ramp	output	[Nmax_bit-1:0]	value of the ramp
o_half_rising_ramp	output		half value of TX ramp

Table 11: Digital Ramp I/O List

5.9.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_116] ASSUMPTION:
The Digital Ramp shall read the i_Nmax_to_reg (N_MAX (DIGITAL_RAMP_CONFIG_PARAM_1 register [3:0]), i_delta_case, i_n_start_case, i_T_delta_to_reg, i_T_preset, and i_T_step1_to_reg from a register set by the user. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_8116] REQUIREMENT:
i_n_start_case shall be lesser or equal than i_Nmax_to_reg. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_2117] REQUIREMENT:
When the PPG_FSM is in the RX_START_UP state, the o_PD_PGA shall go to 0. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_118] REQUIREMENT:
When the input signal o_Tx_preset is set to 1, the o_Tx_Vref shall increment the first step at value i_N_start (DIGITAL_RAMP_CONFIG_PARAM_3 register [2:0]) at a time i_T_preset (DIGITAL_RAMP_CONFIG_PARAM_1 register [5:4]). [END]

[IPOS_STBIO1_MAIN_CONTROLLER_119] REQUIREMENT:

After the first step, the other step shall increase by 1 one every 2-clock cycle. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_5120] TX_PRESET_TIME shall be configurable between 0.5us, 1us, 1.5us and 2us (8,16,24,32 clock cycles) [END]

[IPOS_STBIO1_MAIN_CONTROLLER_120] REQUIREMENT:
When the i_falling_digital_ramp *is set to 1, the* o_TX_Vref *shall decrement from N_MAX value* (DIGITAL_RAMP_CONFIG_PARAM_1 register [3:0]) *to zero with a step equal to* 1 every 2-clock cycle.[END]

[IPOS_STBIO1_MAIN_CONTROLLER_121] REQUIREMENT:
Nmax shall be a value between 1 and 15 configurable by user. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_122] REQUIREMENT:
When o_TX_Vref is equal to i_Nmax_to_reg (N_MAX (DIGITAL_RAMP_CONFIG_PARAM_1 register [3:0])), o_rising_ramp_end shall go high. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_123] REQUIREMENT:
When o_Tx_Vref is equal to 0 during the falling and the rising ramp is ended (o_rising_ramp_end ==0), o_falling_ramp_end shall go high. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_124] REQUIREMENT:
When the rising ramp reaches half its value, o_half_rising_ramp shall go high. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_125] REQUIREMENT:
When i_reset_ramp is high all output shall be down. [END]

5.10. ALC Compensation

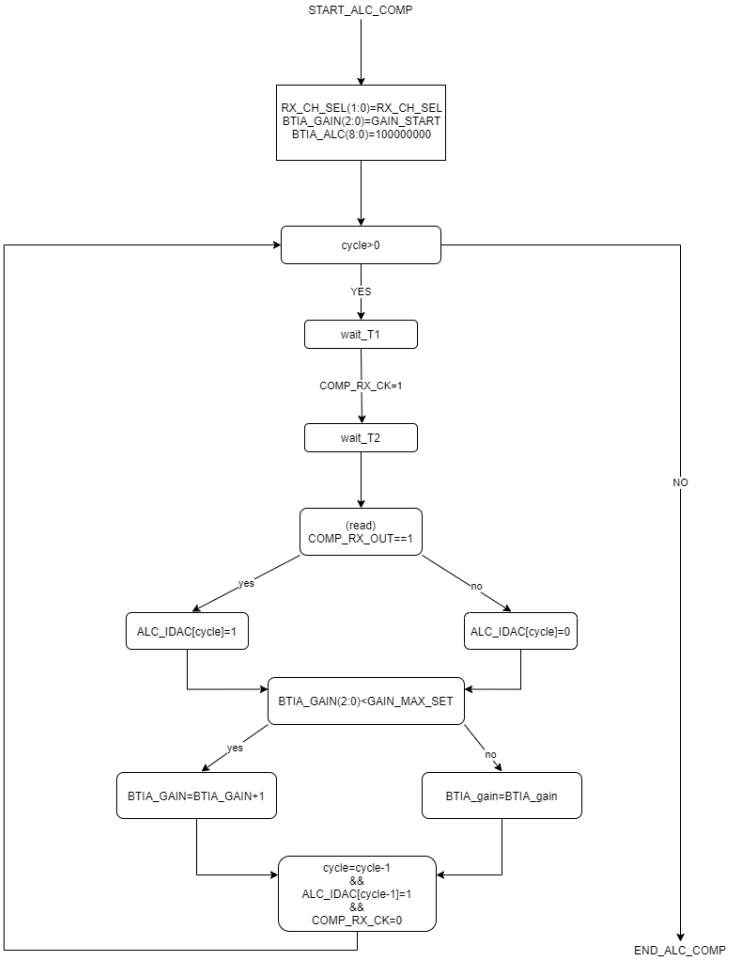


Table 11: ALC Algorithm Flow Diagram

5.10.1. IP Description

This block Is in charge to overseas the Ambient Light Compensation algorithm for the PPG Analog Channel

5.10.2. I/O List

5.10.2.1. Parameters

Generic name	Type	Value	Description
gain_max_bit	int	4	
T1_max_bit_from_decoder	int	4	
T2_max_bit_from_decoder	int	2	
T1_max_bit	int	8	
T2_max_bit	int	5	
BTIA_ALC_register_bit	int	9	
T1_add_T2_max_bit	int	13	
n_iteration_bit	int	4	
ppg_frame_bit	int	4	
ppg_frame_bit_ALC_BTIA	int	12	
n_max_frame	int	5	

5.10.2.2. Ports

Port name	Direction	Type	Description
i_clk	input		16 MHz clock
i_rstn	input		reset signal

Port name	Direction	Type	Description
i_EN_TIA	input		enable tia from ppg fsm
i_EN_BUFF	input		enable buffer from ppg fsm
i_EN_IDAC	input		enable Idac from ppg
i_reset_alc	input		reset alc phase
i_gain_start	input	[gain_max_bit-1:0]	gain start value
i_gain_max_set	input	[gain_max_bit-1:0]	gain max value
i_reset_ppg_start_up	input		reset for start up phase
i_ALC_comp_start	input		start of alc phase
i_comp_rx_out	input		comparator value from analog domain
i_T_TIAON	input	[T1_max_bit_from_decoder-1:0]	TIA Time
i_T1	input	[T1_max_bit_from_decoder-1:0]	T1 value for alc phase
i_T2	input	[T2_max_bit_from_decoder-1:0]	T2 value for alc phase
o_comp_rx_ck	output		comp rx ck for analog domain
o_ALC_comp_end	output		end of alc compensation
o_rx_ch_gain	output	[gain_max_bit-1:0]	rx channel gain

Port name	Direction	Type	Description
o_w_BTIA_ALC	output	[BTIA_ALC_register_bit-1:0]	ALC Corse Result

Table 123: ALC Compensation I/O List

5.10.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_126] REQUIREMENT:
When ALC_COMP starts, o_BTIA_gain (2:0) shall be set at i_gain_start (PPG_ALC_CONFIG_PARAM_2 register [4:2]) and o_w_BTIA_ALC (8:0) shall be set at 100000000. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_127] REQUIREMENT:
ALC cycle algorithm shall be done until the set of the last *o_w_BTIA_ALC* bit. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_728] REQUIREMENT:
At the end of ALC compensation, if ALC is not masked, *o_w_BTIA_ALC relative to the running frame, shall be set to output system (check when i_end_of_compensation rise in PPG FSM). If it's masked, its contribution shall be 0* [END]

[IPOS_STBIO1_MAIN_CONTROLLER_6129] REQUIREMENT:
When *i_ALC_comp_start is set to 1*, the first rise of the o_comp_rx_ck shall happen at least after a T1+T_TIA_On time, both configurable by regmap. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_129] REQUIREMENT:
When *i_ALC_comp_start is set to 1 and the o_BTIA_ALC (ALC Compensation) change is value*, after a configurable time T1 (PPG_ALC_CONFIG_PARAM_1 register [3:0]), *o_comp_rx_ck* shall be high. [END]

[IPOS_STBIO1_MAIN_CONTROLL ER_130] ASSUMPTION:
After T1+T2, if *i_comp_rx_out* is equal to 1, *o_w_BTIA_ALC* [cycle] shall be set to COMP_RX_OUT. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_131] ASSUMPTION:
After T1+T2 *o_w_BTIA_ALC* [cycle - 1] shall be set to 1, until cycle is greater than zero. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_132] REQUIREMENT:
After T1+T2 (PPG_ALC_CONFIG_PARAM_2 register [1:0]), if *o_BTIA_gain* (2:0) < *i_gain_max_set (PPG_ALC_CONFIG_PARAM_2 register [7:5])*, during ALC_COMPENSATION state of PPG fsm with o_end_compensation set to 0, *o_BTIA_gain* increase its value of 1. Instead, if *o_BTIA_gain* 2:0) >= *i_gain_max_set (PPG_ALC_CONFIG_PARAM_2 register [7:5])*, *o_BTIA_gain* shall remain at its old value. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_133] REQUIREMENT:
Once is set to 1, *o_comp_rx_ck* shall be set to 0 after a T2+1 clock cycle. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_9116] REQUIREMENT:
GAIN_START shall be lesser or equal than GAIN_MAX. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_134] REQUIREMENT:
According to time diagram, when in RESET_PPG and ELAB_FSM is in PPG state during the first frame (i_id_frame=0), shall be wait a configurable time PPG_START_UP_TIME (PPG_START_UP_TIME starting from when i_start_operation_ppg is set to 1 and at least 1 clock cycle after i_start_ppg is set to 1 [END]

[IPOS_STBIO1_MAIN_CONTROLLER_7135] REQUIREMENT:
If the ALC_COARSE (o_w_BTIA_ALC) result is equal to zero, it shall be brought to 1[END]



Table 13: ALC Time Diagram

5.11. Average Block

5.11.1. IP Description

This block oversees the “cleaning” of the samples doing simple average math operation and change representation. The output will be 16 bit for ECG, BIA and GSR, 20 bit for PPG instead. For ECG, BIA and GSR the number of acquisitions is fixed, while for PPG is configurable.

5.11.2. I/O List

5.11.2.1. Parameters

Generic name	Type	Value	Description
data_size_bit	int	16	

Generic name	Type	Value	Description
n_average_ecg_bia_gsr_max_bit	int	5	
n_average_ppg_max_bit	int	7	
n_average_ecg_bia_gsr_max	int	17	
n_average_ppg_max	int	128	
max_frame_repetition	int	2	
n_ppg_frame	int	12	
frame_ppg_debug_alc	int	8	

5.11.2.2. Ports

Port name	Direction	Type	Description
i_clk	input		16 MHz Clock
i_rstn	input		Reset
i_data_in	input	[data_size_bit-1:0]	data from adc phases
i_reset_average	input		reset average
i_n_average_ecg_bia_gsr	input	[n_average_ecg_bia_gsr_max_bit-1:0]	number of acquisition for ECG/BIA/GSR
i_reset_average_ppg	input		reset average
i_frame_ppg_debug	input	[n_ppg_frame_bit-1:0]	frame selection for ppg data debug

Port name	Direction	Type	Description
i_force_reset	input		force reset avg
i_save_alc	input		save alc value in regmap
i_ppg_ioff_calib	input		ppg ioff calibration phase
i_alc_mask	input		mask alc average result
i_ppg_ioff_data_calib_n_m0	input		nm0 computation
i_ppg_ioff_data_calib_n_m1	input		nm1 computation
i_reset_avg_calib_0	input		reset average for nm0
i_reset_avg_calib_1	input		reset average for nm1
i_id_frame	input	[n_ppg_frame_bit-1:0]	
i_gsr_result	input		start computation for gsr result
i_en_gsr_cds	input		enabling gsr cds mode
i_n_average_ppg	input	[n_average_ppg_max_bit-1:0]	number of acquisition for ppg
i_data_valid_ecg_bia_adc	input		data valid for ecg and bia data
i_data_valid_ppg_adc	input		data valid for ppg data
i_data_valid_gsr_adc	input		data valid for gsr
i_data_valid_noise_ppg_adc	input		data valid for noise ppg
i_end_of_conversion	input		end of conversion from adc phases

Port name	Direction	Type	Description
i_average_repetion	input	[max_frame_repetition-1:0]	number of ppg frame repetition
o_data_out	output	[data_size_out_bit-1:0]	output data after average
o_end_average	output		end average operation
o_alc_data	output	[data_size_out_bit+2:0]	alc data output
o_ppg_data_raw	output	[data_size_out_bit+2:0]	ppg data raw output
o_n_m0	output	[data_size_bit-1:0]	nm0 data output
o_n_m1	output	[data_size_bit-1:0]	nm1 data output
o_saturation_flag	output	[n_ppg_frame-1:0]	saturation flag for ppg
o_adc_data_debug_alc_1	output	[frame_ppg_debug_alc-1:0]	alc 1 registers data debug
o_adc_data_debug_alc_2	output	[frame_ppg_debug_alc-1:0]	alc 2 registers data debug
o_adc_data_debug_ppg_raw	output	[frame_ppg_debug_ppg_raw-1:0]	ppg raw registers data debug
o_write_enable	output		start ahb operation

Table 145: Average Block I/O List

5.11.3. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_136] ASSUMPTION:
The Average block shall accumulate data every time the i_end_of_conversion signal is asserted. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1136] REQUIREMENT:
After the assertion of i_end_of_conversion a number of times equals to i_n_average_ecg_bia_gsr (in case of ECG, BIA and GSR) or i_n_average_ppg (in case of PPG without frame repetitions), after at least one clock cycle o_data_result shall be computed and o_write_enable shall rise [END]

[IPOS_STBIO1_MAIN_CONTROLLER_137] REQUIREMENT:
The average block shall perform an average on 4 Samples for each of ECG_x_AC, BIA AC P, BIA AC Q, BIA DC P and BIA DC Q, 1 Sample ECG_x_DC (Check if the i_end_of_coversion rise the correct number of times) [END]

[IPOS_STBIO1_MAIN_CONTROLLER_138] REQUIREMENT:
The average block shall perform the average on 16 sample acquisition for GSR. When the CDS mode is enabled, the average block shall perform an average on 16 sample acquisition on GSR ON and 16 sample acquisition on GSR OFF, then shall perform a subtraction operation between the two results. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1139] REQUIREMENT:
The PPG Noise Data shall be accumulated in two phases, the first when the PPG FSM state is in FIRST_ALC_SAMPLES States and the second when the PPG FSM state is in the SECOND_ALC_SAMPLES [END]

[IPOS_STBIO1_MAIN_CONTROLLER_139] REQUIREMENT:
When a PPG Raw data (i_data_valid_ppg_adc set to 1) or a PPG_Noise (i_data_valid_noise_ppg_adc set to 1) data is sampled (i_end_of_conversion set to 1) the data shall be accumulated until i_n_average_ppg is not reached. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_140] REQUIREMENT:
Once the PPG Raw Data and PPG Noise reach the i_n_average_ppg value, depending on the ALC Mask bit value, the two accumulated shall be subtracted and final results shall be computed as follow:

I_N_Average_PPG	Samples	Result with ALC MASK=0	Result with ALC Mask=1
0	3= (N1+P+N2)	(Acc_P-(Acc_N)/2) * 8	Acc_P*16
2	4= (N1+2P+N2)	(Acc_P -(Acc_N)) * 4	Acc_P *8
4	8= (2N1+4P+2N2)	(Acc_P -(Acc_N)) * 2	Acc_P *4
8	16= (4N1+8P+4N2)	(Acc_P -(Acc_N))	Acc_P *2
16	32= (8N1+16P+8N2)	(Acc_P -(Acc_N))/2	Acc_P
32	64= (16N1+32P+16N2)	(Acc_P -(Acc_N))/4	Acc_P /2
64	128= (32N1+64P+32N2)	(Acc_P -(Acc_N))/8	Acc_P /4
128	256= (64N1+128P+64N2)	(Acc_P -(Acc_N))/16	Acc_P /8

Table 18: PPG Result Table

Where PPG Raw Data is indicated as P and PPG Noise is indicated as N1 and N2, while Acc_P is the accumulated PPG data and Acc_N is the accumulated noise[END]

[IPOS_STBIO1_MAIN_CONTROLLER_9139] REQUIREMENT:
If during the PPG Frame computation, the i_alc_mask (ALC Disabled) is enabled, the final result shall be only the ppg acquisition and the subtraction of the alc data shall not be performed. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_143] REQUIREMENT:
When i_end_of_conversion is set to 1 and i_n_average_ecg_bia_gsr (average ecg,bia and gsr) or i_n_average_ppg (Frame_X_Parameter_1) is equal to their limit values, an average operation shall be done and o_write_enable shall be set to 1. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_1143] REQUIREMENT:
For GSR Operation, if i_en_cds is set 0, the behavior shall as described in Requirement 143, otherwise the o_write_enabled shall be asserted when i_end_of_coversion is set to 1 and i_n_average_ecg_bia_gsr is equal to their limit values when the measurement is done with o_gsr_curr_on set to 0 and o_gsr_on set to 1[END]

[IPOS_STBIO1_MAIN_CONTROLLER_2143] REQUIREMENT:
When i_frame_ppg_debug is set to a value greater than 0 and i_n_average_ppg is set to a value between 1 and 16, all the ADC output (i_data_in) for ALC (i_data_valid_noise_ppg_adc set to 1) and PPG data (i_data_valid_ppg_adc) shall be stored in o_adc_data_debug_alc_2, o_adc_data_debug_alc_1 and o_adc_data_debug_ppg_raw. [END]

[IPOS_STBIO1_MAIN_CONTROLLER_2144] REQUIREMENT:
[Covers: DDS_STBIO1_0124]
[Covers: DDS_STBIO1_0142]
The data written in FIFO shall have the following tag:

DATA TYPE	BIT: 23 - 20	BIT: 19 - 16	BIT: 15 - 0
PPG FRAME 0	0000	DATA 20 BIT	
PPG FRAME 1	0001	DATA 20 BIT	
PPG FRAME 2	0010	DATA 20 BIT	
PPG FRAME 3	0011	DATA 20 BIT	
PPG FRAME 4	0100	DATA 20 BIT	
PPG FRAME 5	0101	DATA 20 BIT	
PPG FRAME 6	0110	DATA 20 BIT	
PPG FRAME 7	0111	DATA 20 BIT	
PPG FRAME 8	1000	DATA 20 BIT	
PPG FRAME 9	1001	DATA 20 BIT	
PPG FRAME 10	1010	DATA 20 BIT	
PPG FRAME 11	1011	DATA 20 BIT	
TIME SLOT DATA	1111	0000	DATA 16 BIT
ECG0_AC	1100	0000	DATA 16 BIT
ECG0_DC	1101	0000	DATA 16 BIT
ECG1_AC	1100	0001	DATA 16 BIT
ECG1_DC	1101	0001	DATA 16 BIT
ECG2_AC	1100	0010	DATA 16 BIT
ECG2_DC	1101	0010	DATA 16 BIT

BIA_AC_P	1110	0001	DATA 16 BIT
BIA_DC_P	1110	0011	DATA 16 BIT

Table 2: DATA TAG Table

[END]

[IPOS_STBIO1_MAIN_CONTROLLER_2146] REQUIREMENT:
In case the ppg sample data are equal (once converted in two's complement) at 0x8000 or 0x7FFF for at least half of the selected average (example, if 128 acquisitions are selected, the threshold shall be greater or equal to 64), o_saturation_flag shall be set to 1 and shall go to 0 at the start of the next frame or frame repetition.
[END]

5.11.4. PPG Frame Repetition

The frame repetition is a classic mean operation between two 20 bit PPG data. The result of the frame is accumulated on a 24 bit accumulator that is divided by the number of repeated frame. The number of repetition frames supported is 0 (no frame repetead), 2,4 and 8.

5.11.4.1. Requirements

[IPOS_STBIO1_MAIN_CONTROLLER_2145] REQUIREMENT:
The Frame shall work as follows:

1. Select how many times a particular frame shall be repeated
2. The result of each repetition shall be obtained as explained in **[IPOS_STBIO1_MAIN_CONTROLLER_140] REQUIREMENT**
3. After the last repetition has been accumulated, the data output shall be obtained dividing by the number of repeated frames the accumulated data to obtain a 20 bit data.

The final result shall be computed as follows:

Number of times the Frame has been repeated	Result
2	(Res_F_R1 + Res_F_R2)/2
4	(Res_F_R1 + Res_F_R2+Res_F_R3+Res_F_R4)/4
8	(Res_F_R1 + Res_F_R2+Res_F_R3+…..+Res_F_R8)/8

Table 20: Frame Repetitions Result Computation table

Where Res_F_Rx stands for Result_frame_Repeated_times (Example: RES_F_R2 refers to the result of the second repetition of the frame.
[END]

6. AHB Write Transaction

6.1. Write Transaction

Main Controller only supports single write transaction at 8/32 bit.


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7. Appendix

7.1. Example of Results for PPG IOFF Calibration

BGAIN	RGAIN	Nstep1	Nm0	Nm1	Nstep2
0	8.00E+3	55.00	32768	26603	267
1	16.00E+3	27.00	32768	26715	133
2	32.00E+3	13.00	32768	26939	66
3	64.00E+3	7.00	32768	26491	33
4	128.00E+3	3.00	32768	27388	16
5	256.00E+3	1.00	32768	29181	8
6	512.00E+3	1.00	32768	25594	4
7	1.024E+6	1.00	32768	18421	2
8	2.048E+6	1.00	32768	4073	1

Figure 16: PPG IOFF Calibration example Results

7.2. AHB Protocol Reference

https://developer.arm.com/documentation/ihi0033/latest/